

Chapter 12 :

Multistage Amplifiers

12.1 Introduction

In some electronic systems, the input signal voltage is very small and the output load needs a very large voltage, for example, an amplifier system of open-air concerts. The microphone produces a small voltage signal (mV) and the loud speakers need a high power (high voltage and current). In such systems, a single-stage amplifier cannot produce all of the required voltage (or power) gain. In these systems, the signal is amplified in multiple stages (amplifiers). The first stage that amplifies the voltage levels of the signals from transducers (like microphones, antenna, heads of HDD etc.) is called **preamplifier**. After the preamplifier, there are one or two amplifiers that also amplify the voltage levels of the signals are called **intermediate stage** (or amplifier). The last stage, to which the load is connected, mainly amplifies the current levels (i.e. power) of the signals is called **power amplifier**. As this stage drives the output load, it is also called **driver amplifier** (or simply **driver**). Thus, a complete electronic system may consist of a number of amplifier stages: each consisting of a number of resistors, capacitors and transistors (or IC). Amplifiers that provide voltage, current and/or power gain by using two more stages of amplifiers is called a **multistage amplifier**. So far we have studied how to analyze and design a single stage amplifier using BJTs and FETs. In this chapter, we will study all about multistage voltage amplifiers. The power amplifiers will be covered in Chapter 15 [H_C_atten].

12.2 Voltage Gain of a Multistage Amplifier

When the output of one amplifier is connected to the input of next amplifier, the arrangement is called a **cascaded amplifier**. Generally most of the multistage amplifiers are cascaded amplifiers. Fig.12-1 shows a multistage amplifier that consists of two voltage amplifiers connected in cascaded form. To explain how the overall voltage gain of this multistage amplifier is calculated, let us assume that the voltage gain of the 1st stage and 2nd stage are A_{v1} , and A_{v2} , respectively. The input voltage of the 1st stage is v_{i1} . Therefore, the output voltage of the 1st stage will be $v_{o1} = v_{i1} \times A_{v1}$. As the output of 1st stage is connected to the 2nd stage, so the input of the 2nd stage will be $v_{i2} = v_{o1} = v_{i1} \times A_{v1}$. This input voltage will be amplified again by the 2nd amplifier. So the output of the 2nd amplifier will be $v_{o2} = v_{i2} \times A_{v2} = (v_{i1} \times A_{v1})A_{v2}$. Therefore, the overall voltage gain of this multistage amplifier will be,

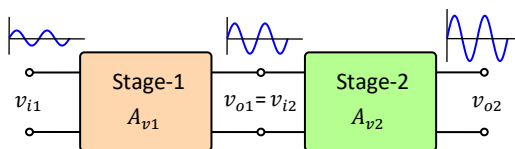


Fig.12-1: Two-stage cascaded amplifier

$$A_{vm} = \frac{v_{o2}}{v_{i1}} = \frac{(v_{i1} \times A_{v1})A_{v2}}{v_{i1}} = A_{v1} \times A_{v2} \quad (12-1)$$

From this expression, we find that the overall gain of a two-stage cascaded amplifier is simply the product of the gain of individual amplifiers.

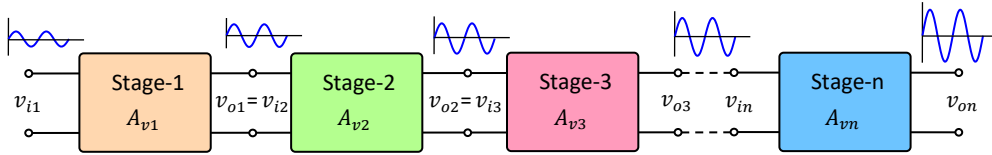


Fig.12-2: N-stage cascaded amplifier

Now, instead of two stages, if we cascade N number of stages, as shown in Fig.12-2, the overall gain of this multistage amplifier will be simply,

$$A_v = A_{v1} \times A_{v2} \times A_{v3} \dots \times A_{vn} \quad (12-2)$$

If the gain of every stage is equal, that is, $A_{v1} = A_{v2} = A_{v3} \dots = A_{vn} = A_v$, then the overall gain will be,

$$A_{vm} = A_v^n \quad (12-3)$$

In a multistage amplifier some of the stages (or all) may have 180° phase shift (like CE amplifiers). The gain of these stages will be a negative number. In calculating the overall gain of the multistage amplifier we may only use the gain not the sign! However, if we consider the sign of the gain that is if we use Equ.(12-2) algebraically, we will get the overall phase shift of the multistage amplifier. If Equ.(12-2) gives a positive product, it means that the multistage amplifier has no phase shift, and if the Equ.(12-2) gives a negative value, then the multistage amplifier will have 180° phase shift.

To find the overall voltage gain of the multistage amplifier in dB unit, we will ignore the algebraic sign of voltage gain of the whole amplifier as well as its individual stages. According to the definition, the voltage gain of the multistage amplifier in dB, can be calculated using Equ.(12-2) as,

$$A_{vm}(\text{dB}) = 20 \log_{10} A_{vm} = 20 \log_{10} (A_{v1} \times A_{v2} \times A_{v3} \dots \times A_{vn}) \text{ dB}$$

$$\text{or,} \quad A_{vm}(\text{dB}) = 20 \log_{10} A_{v1} + 20 \log_{10} A_{v2} + \dots + 20 \log_{10} A_{vn}$$

$$\text{or,} \quad A_{vm}(\text{dB}) = A_{v1}(\text{dB}) + A_{v2}(\text{dB}) + \dots + A_{vn}(\text{dB}) \quad (12-4)$$

From Equ.(12-4), we find that the overall voltage gain of the multistage amplifier in dB unit is the sum of the voltage gain in dB of individual stages.

Equ.(12-2) to Equ.(12-4) are derived for voltage gain, but they can also be used for current gain and power gain. In case of power, the gain in dB is calculated as $P(\text{dB}) = 10 \log_{10} P$. Moreover, here the loading effect of different stages and the source internal resistance have not been considered, that is A_{v1} to A_{vn} are the voltage gain when they are connected in the circuit (after being affected by the internal resistance of the amplifiers). These gains are called **in-circuit voltage gain**.

Example 12-1

Suppose that the multistage amplifier of Fig.12-2 consists of five stages with gain of $A_{v1} = 13$, $A_{v2} = -15$, $A_{v3} = 17$, $A_{v4} = -12$, and $A_{v5} = 20$. Calculate the overall gain of the cascaded amplifier and its phase. Also calculate the overall gain in dB.

Solution:

From Equ.(12-2) $A_{vm} = A_{v1} \times A_{v2} \times A_{v3} \times A_{v4} \times A_{v5}$

or, $A_{vm} = 13 \times (-15) \times 17 \times (-12) \times 20 = 795,600$ [Ans.]

As the overall gain is positive, the multistage amplifier will have 0° phase shift.

Gain in dB, $A_{vm}(\text{dB}) = 20 \log_{10}(795600) \approx 118 \text{ dB}$ [Ans.]

Comments: A very large absolute voltage gain can be represented by a small number using dB units.

12.3 Loading Effect in Multistage Amplifiers

Already we have mentioned that Equ.(12-1) to Equ.(12-4) have been derived using the in-circuit voltage gain of individual stages. That is, A_{v1} to A_{vn} are the values of voltage gain when the corresponding units are connected in cascade. Thus, we have assumed that each value of individual stage gain has already taken into account the loading effect on the output of a stage by the input of the next stage. However, the effect of source internal resistance (r_s) at the first stage, and the effect of the load resistance (R_L) at the last stage has not been considered. If the *open-circuit* (unloaded) voltage gains of the individual stage are given, then these equations will not work. Then, we have to calculate the overall voltage gain of the multistage amplifier by taking into account the loading effect of individual stages. To do that, we must know the internal input and output resistances of the amplifiers. Moreover, to take into account the effect of r_s and R_L , these values must also be known. Theoretically, the loading effect present on a stage may depend on all of the successive stages, due to some feedback effect. We will assume that the values of input and output impedances are calculated considering that feedback effects.

To calculate the overall voltage gain taking into the loading effect, we will take a three-stage amplifier as shown in Fig.12-3. Here, the gain A_{v1} to A_{v3} are the open-circuit voltage gains. That is, A_{v1} is the voltage gain of the 1st stage, when the 2nd stage is not connected at its output terminals and so on.

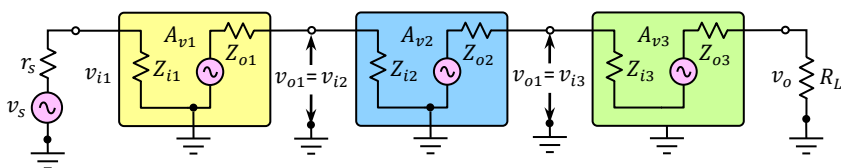


Fig.12-3: Three-stage cascaded amplifier

From this figure, the value of voltages at different stages can be calculated simply using voltage divider rule,

$$v_{i1} = \frac{Z_{i1}}{r_s + Z_{i1}} v_s$$

$$v_{o1} = v_{i2} = \frac{Z_{i2}}{Z_{o1} + Z_{i2}} A_{v1} v_{i1}$$

$$v_{o2} = v_{i3} = \frac{Z_{i3}}{Z_{o2} + Z_{i3}} A_{v2} v_{i2}$$

$$v_o = \frac{R_L}{Z_{o3} + R_L} A_{v3} v_{i3}$$

Therefore, the overall gain of this multistage amplifier can be found by combining these equations as,

$$v_o = \frac{R_L}{Z_{o3} + R_L} A_{v3} \times \frac{Z_{i3}}{Z_{o2} + Z_{i3}} A_{v2} \times \frac{Z_{i2}}{Z_{o1} + Z_{i2}} A_{v1} \times \frac{Z_{i1}}{r_s + Z_{i1}} v_s$$

or,

$$A_{vm} = \frac{v_o}{v_s} = \frac{(Z_{i1} Z_{i2} Z_{i3} R_L)(A_{v1} A_{v2} A_{v3})}{(r_s + Z_{i1})(Z_{o1} + Z_{i2})(Z_{o2} + Z_{i3})(Z_{o3} + R_L)} \quad (12-5)$$

From Equ.(12-5), we find that the overall voltage gain of a multistage amplifier is the product of the open-circuit voltage gains of individual stages, and the voltage division ratios due to the loading effects of one another.

Example 12-2

Suppose a multistage amplifier is constructed cascading three small signal amplifiers. The input impedances, open-circuit voltage gains, and output impedances of the amplifiers are, $Z_{i1} = 1.2 \text{ k}$, $A_{v1} = 20$, $Z_{o1} = 3 \text{ k}$, $Z_{i2} = 3.3 \text{ k}$, $A_{v2} = 15$, $Z_{o2} = 1.5 \text{ k}$, and $Z_{i3} = 2.5 \text{ k}$, $A_{v3} = 12$, $Z_{o3} = 1.2 \text{ k}$. The source internal resistor is 200Ω and the load resistance is 2.2 k . Calculate the value of the voltage produced across the load for the source voltage of $v_s = 20 \text{ mV (RMS)}$.

Solution:

Here, $r_s = 200 \Omega$, and $R_L = 2.2 \text{ k}$. Using Equ.(12-5), we will calculate the overall gain of the multistage amplifier as,

$$A_{vm} = \frac{v_o}{v_s} = \frac{(Z_{i1} Z_{i2} Z_{i3} R_L)(A_{v1} A_{v2} A_{v3})}{(r_s + Z_{i1})(Z_{o1} + Z_{i2})(Z_{o2} + Z_{i3})(Z_{o3} + R_L)}$$

or,

$$A_{vm} = \frac{v_o}{v_s} = \frac{(1.2 \text{ k} \times 3.3 \text{ k} \times 2.5 \text{ k} \times 2.2 \text{ k})(20 \times 15 \times 12)}{(200 \Omega + 1.2 \text{ k})(3 \text{ k} + 3.3 \text{ k})(1.5 \text{ k} + 2.5 \text{ k})(1.2 \text{ k} + 2.2 \text{ k})} \approx 654 \text{ [Ans.]}$$

As the source voltage is $v_s = 20 \text{ mV (RMS)}$, the output voltage will be,

$$v_o = 654 \times v_s = 654 \times 20 \text{ mV (RMS)} = 13.1 \text{ V (RMS)} \text{ [Ans.]}$$

Comments: Although the product of the open-circuit gains is $20 \times 15 \times 12 = 3600$, due to the loading effect the overall gain has been only 654.



12.4 Frequency Response of Multistage Amplifiers

If we develop a multistage amplifier by cascading two or more amplifier stages, then every stage will have its own lower and higher cut-off frequencies. If the lower cut-off frequencies of the individual amplifiers are far apart (at least one decade) in values, we can consider the highest one as the overall lower cut-off frequency of the multistage amplifier. Similarly, if the higher cut-off frequencies of the individual stages are far apart (at least one decade) in values, we can consider the lowest one as the overall higher cut-off frequency of the multistage amplifier. However, generally in a multistage amplifier the individual stages have identical lower cut-off and upper cut-off frequencies. In that case, the overall higher and lower cut-off frequencies of the multistage amplifier have to be calculated using the following expressions:

$$f_{L(m)} = \frac{f_L}{\sqrt{2^{1/n} - 1}} \quad (12-6)$$

and,
$$f_{H(m)} = f_H \sqrt{2^{1/n} - 1} \quad (12-7)$$

where, $f_{L(m)}$ = overall lower cut-off frequency of the multistage amplifier

$f_{H(m)}$ = overall higher cut-off frequency of the multistage amplifier

f_L = lower cut-off frequency of individual stages (as $f_{L1} = f_{L2} = \dots f_L$)

f_H = higher cut-off frequency of individual stages (as $f_{H1} = f_{H2} = \dots f_H$)

n = number of stages having identical lower and higher cut-off frequencies used to make the multistage amplifier.

From Equ.(12-6) we can find that (see Example 12-3) the overall lower cut-off frequency of multistage amplifier is higher than that of individual amplifier, and from Equ.(12-7) the higher cut-off frequency is lower than that of individual amplifier. Therefore, the bandwidth of a multistage amplifier will be lower than the bandwidth of the individual amplifiers.

If we want to design a multistage amplifier with a particular lower and a higher cut-off frequency, then we can design the individual stages with identical lower and higher cut-off frequencies determined as follows:

$$f_L = f_{L(m)} \sqrt{2^{1/n} - 1} \quad (12-8)$$

and,
$$f_H = \frac{f_{H(m)}}{\sqrt{2^{1/n} - 1}} \quad (12-9)$$

Example 12-3

The lower and higher cut-off frequencies of a single stage CE amplifier are 100 Hz, and 1 MHz, respectively. A multistage amplifier is made by cascading three such CE amplifiers. Calculate the overall lower and higher cut-off frequencies of the multistage amplifier.

Solution:

As the three stages of the multistage amplifier have identical lower and higher cut-off frequencies, the overall cut-off frequencies have to be calculated using Equ.(12-6) and Equ.(12-7).

Using Equ.(12-6),
$$f_{L(m)} = \frac{f_L}{\sqrt{2^{1/n} - 1}} = \frac{100 \text{ Hz}}{\sqrt{2^{1/3} - 1}} \approx 196 \text{ Hz [Ans.]}$$

Using Equ.(12-7),
$$f_{H(m)} = f_H \sqrt{2^{1/n} - 1} = 1 \text{ MHz} \sqrt{2^{1/3} - 1} \approx 510 \text{ kHz [Ans.]}$$

Comments: The overall lower cut-off frequency of multistage amplifier is higher than that of individual amplifier, and the higher cut-off frequency is lower than that of individual amplifier.

12.5 Coupling Methods of Multistage Amplifiers

The circuitry used to connect the output of one stage of a multistage amplifier to the output of the next stage amplifier is called **coupling method**. In the single stage amplifiers, discussed so far, we have used RC (resistor-capacitor) coupling method to connect the sources and the loads to the amplifiers. This RC coupling method is also used for multistage amplifiers. Another two methods used for connecting multistage amplifiers are: **direct coupling**, and **transformer coupling**.

12.6 RC Coupling Method

We have already discussed that the primary reason for using RC coupling is to block the DC voltage levels of the amplifier from passing through the source and loads. In multistage amplifier RC coupling is also used to prevent the DC voltage level of output of one stage to be applied to the input of the next stage. The capacitor between two stages only allows the AC signal to flow from one stage to another. The main advantage of RC coupling is: we can have different DC bias voltage levels at the output of one stage and the input of the next stage. This idea is illustrated in Fig.12-4, which shows the RC coupling method of two amplifiers.

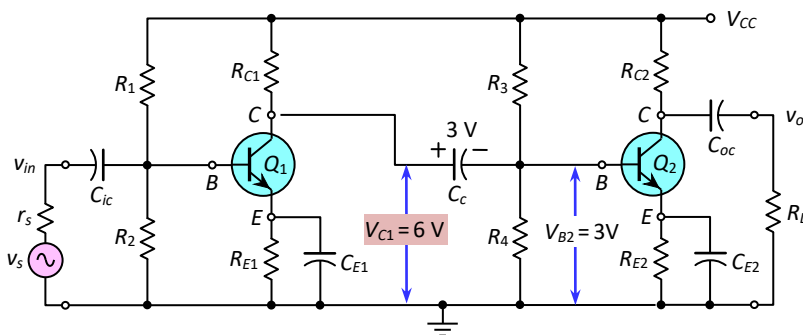


Fig.12-4: Two-stage RC coupled CE amplifier

The output of the first CE amplifier is connected to the input of next CE amplifier through the coupling capacitor C_c . The source is capacitor-coupled to the input of first stage, and the load is capacitor-coupled to the output of last stage. The signal will be amplified by each stage, and each stage will produce 180° phase shift. Therefore, the overall phase shift will be 360° or 0° . As shown in this figure the DC voltage level of the collector of first stage is 6 V and the DC voltage level of the base of second stage is 3 V. So the DC voltage drop across the capacitor is $6 \text{ V} - 3 \text{ V} = 3 \text{ V}$. The capacitor must have the tolerance voltage greater than this voltage drop (3 V in this case). The

reactance of the coupling capacitor (X_{C_c}) should be low enough to pass the AC signal of desired lowest frequency from first stage to the second stage. The disadvantage of RC coupling method is that it affects the low frequency response of the multistage amplifier. RC coupling sometimes results in unreasonably large coupling capacitors and/or impractically large lower cut-off frequency. RC coupling is not used in integrated circuits, because it is difficult to fabricate capacitor of required size in ICs. Direct coupling method can eliminate these problems.

12.6.1 AC Analysis of RC Coupled Amplifiers

AC analysis of RC coupled multistage amplifier is similar to the analysis of single-stage amplifiers. We can analyze a multistage amplifier using r_e model or hybrid model of transistor as described in Chapter 6. The hybrid equivalent circuit of the two-stage amplifier of Fig.12-4 is shown in Fig.12-5. To get the mid-band frequency response, all external capacitors have been considered short and all parasitic capacitances are considered open. To get the AC response, the DC voltage sources have also been considered short.

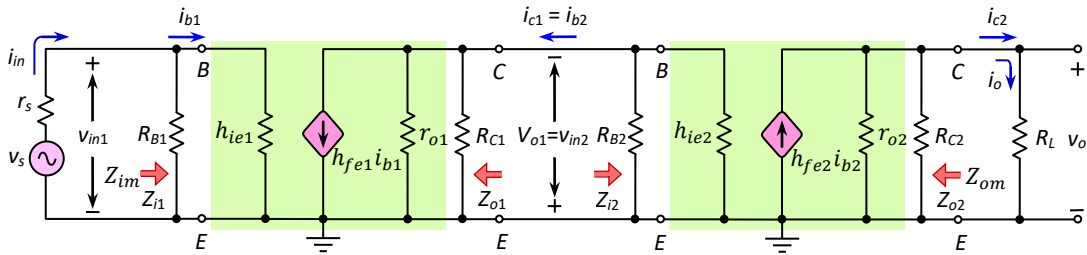


Fig.12-5: AC equivalent circuit of two-stage RC coupled CE amplifier

The AC equivalent circuit consists of cascading AC equivalent circuits of two CE amplifiers. As before, voltage polarities and current directions are indicated for a positive instantaneous value of AC signal. Look at the polarity of output voltage: due to 180° phase shift of each stage, the final output (v_o) is in phase with input (v_{in1}). From this equivalent circuit and the concept of AC analysis described in Chapter 6, the required expressions of this amplifier can easily be determined. The input, output impedance of this amplifier will be,

$$Z_{im} = Z_{i1} = R_{B1} \parallel h_{ie1} \quad (12-10)$$

$$\text{and,} \quad Z_{om} = Z_{o2} = r_{o2} \parallel R_{C2} \approx R_{C2} \quad (12-11)$$

Input impedance of the second stage will be,

$$Z_{i2} = R_{B2} \parallel h_{ie2} \quad (12-12)$$

Voltage gain of the first stage will be,

$$A_{v1} = -\frac{h_{fe1}(r_{o1} \parallel R_{C1} \parallel Z_{i2})}{h_{ie1}} = -\frac{h_{fe1}(r_{o1} \parallel R_{C1} \parallel R_{B2} \parallel h_{ie2})}{h_{ie1}} \quad (12-13)$$

$$\because r_o \text{ is very large,} \quad A_{v1} \approx -\frac{h_{fe1}(R_{C1} \parallel R_{B2} \parallel h_{ie2})}{h_{ie1}} \quad (12-14)$$

Voltage gain of the second stage will be,

$$A_{v2} = -\frac{h_{fe2}(r_{o2} \parallel R_{C2} \parallel R_L)}{h_{ie2}}$$

$\because r_o$ is very large,

$$A_{v2} \approx -\frac{h_{fe2}(R_{C2} \parallel R_L)}{h_{ie2}} \quad (12-15)$$

The overall voltage gain of the multistage amplifier will be the product of the individual voltage gain. Therefore,

$$A_{vm} \approx A_{v1} \times A_{v2} = \frac{h_{fe1}(R_{C1} \parallel R_{B2} \parallel h_{ie2})}{h_{ie1}} \times \frac{h_{fe2}(R_{C2} \parallel R_L)}{h_{ie2}} \quad (12-16)$$

12.6.2 Design of RC Coupled Multistage CE Amplifiers

The simplest approach to design an RC coupled multistage amplifier is to make each stage identical. At first, we have to design a single stage amplifier using the process discussed in Chapter 10, then cascade two or three (or even more) of them. Let us consider a two stage amplifier of Fig.12-4. Values of all resistors can be calculated using equations given in Chapter 10. Then, we have to use $R_1 = R_3$, $R_2 = R_4$, $R_{C1} = R_{C2}$, and $R_{E1} = R_{E2}$. Values of C_{ic} and C_{oc} can be calculated using Equ.(10-8), and Equ.(10-9), respectively (of Chapter 10). The value of C_C can be calculated using Equ.(10-9) with a little change as,

$$C_C = \frac{10}{2\pi f_L(Z_{o1} + Z_{i2})} \approx \frac{10}{2\pi f_L(R_{C1} + R_{B2} \parallel h_{ie2})} \quad (12-17)$$

where, Z_{o1} is the output impedance of the first stage and Z_{i2} is the input impedance of second stage.

We know that the lower cut-off frequency of a CE amplifier is determined by the emitter bypass capacitor C_E . So its value has to be calculated more accurately. For a required lower cut-off frequency of the multistage amplifier ($f_{L(m)}$), the lower cut-off frequency (f_L) of individual stage should a little lower than $f_{L(m)}$. The value of f_L can be calculated using Equ.(12-8). For a two stage amplifier of Fig.(12-4), the value of $n = 2$. Therefore, using Equ.(12-8),

$$f_L = f_{L(m)} \sqrt{2^{1/n} - 1} = f_{L(m)} \sqrt{2^{1/2} - 1} \approx 0.65 f_{L(m)} \quad (12-18)$$

Now, the value of C_{E1} has to be calculated using Equ.(10-7) as,

$$C_{E1} = \frac{1 + h_{fe}}{2\pi f_L h_{ie}} = \frac{1 + h_{fe}}{2\pi [0.65 f_{L(m)}] h_{ie}} \quad (12-19)$$

Finally, we have to use $C_{E2} = C_{E1}$.

Example 12-4

Fig.12-6 shows a multistage CE amplifier with voltage divider bias. The transistors, used in this circuit, have $h_{ie} = 1.3 \text{ k}$, $h_{fe} = 100$, and $r_o = 100 \text{ k}$. Calculate A_{v1} , A_{v2} , Z_{im} , A_{vm} , and Z_{om} . Calculate the voltage gain and lower cut-off frequency of individual amplifiers and those of the multistage amplifier.

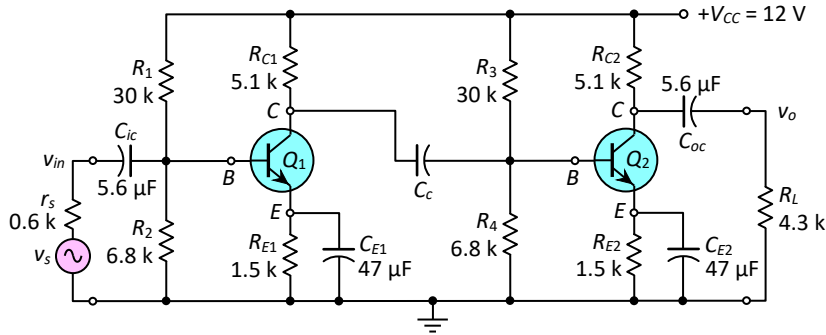


Fig.12-6: Two-stage RC coupled amplifier for Example 12-4

Solution:

Value of bias resistance, $R_{B1} = R_1 \parallel R_2 = \frac{30 \text{ k} \times 6.8 \text{ k}}{30 \text{ k} + 6.8 \text{ k}} \approx 5.54 \text{ k} = R_{B2}$

Value of AC load resistance, $R'_L = R_{C2} \parallel R_L = \frac{R_C R_L}{R_C + R_L} = \frac{5.1 \text{ k} \times 4.3 \text{ k}}{5.1 \text{ k} + 4.3 \text{ k}} \approx 2.3 \text{ k}$

Value of r_o , $r_o = \frac{1}{h_{oe}} = \frac{1}{10 \times 10^{-6} \text{ S}} = 100 \text{ k}\Omega$

Using Equ.(12-10), $Z_{i1} = R_{B1} \parallel h_{ie1} = \frac{5.54 \text{ k} \times 1.3 \text{ k}}{5.54 \text{ k} + 1.3 \text{ k}} \approx 1 \text{ k}\Omega = Z_{i2}$

Using Equ.(12-60), $Z_{o1} = r_{o1} \parallel R_{C1} = \frac{100 \text{ k} \times 5.1 \text{ k}}{100 \text{ k} + 5.1 \text{ k}} \approx 4.85 \text{ k}\Omega$

Voltage gain can be calculated using Equ.(12-13).

For 1st stage, $A_{v1} = -\frac{h_{fe1}(r_{o1} \parallel R_{C1} \parallel Z_{i2})}{h_{ie}} = -\frac{100(100 \text{ k} \parallel 5.1 \text{ k} \parallel 1 \text{ k})}{1.3 \text{ k}\Omega} \approx -64 \text{ [Ans.]}$

For 2nd stage, $A_{v2} = -\frac{h_{fe2}(r_{o2} \parallel R_{C2} \parallel R_L)}{h_{ie}} = -\frac{100(100 \text{ k} \parallel 5.1 \text{ k} \parallel 4.3 \text{ k})}{1.3 \text{ k}\Omega} \approx -175 \text{ [Ans.]}$

The overall voltage gain of the two-stage amplifier will be,

$$A_{vm} = A_{v1} \times A_{v2} = -64 \times -175 = 11200 \text{ [Ans.]}$$

As the lower cut-off frequency of CE amplifier is set by emitter bypass capacitor, we can consider that the lower cut-off frequencies for all other capacitors will be very small. Let us check.

Using Equ.(10-8), $f_{L(C_{ic})} = \frac{1}{2\pi(r_s + Z_i)C_{ic}} = \frac{1}{2\pi(r_s + R_{B1} \parallel h_{ie1})C_{ic}}$

or, $f_{L(C_{ic})} = \frac{1}{2\pi(0.6 \text{ k} + 5.54 \text{ k} \parallel 1.3 \text{ k}) \times 5.6 \mu\text{F}} \approx 17 \text{ Hz}$

Using Equ.(10-9)
$$f_{L(C_{oc})} = \frac{1}{2\pi(Z_o + R_L)C_{oc}} = \frac{1}{2\pi(r_o \parallel R_{C2} + R_L)C_{oc}}$$

or,
$$f_{L(C_{oc})} = \frac{1}{2\pi(100 \text{ k} \parallel 5.54 \text{ k} + 4.3 \text{ k}) \times 5.6 \text{ } \mu\text{F}} \approx 19 \text{ Hz}$$

The lower cut-off frequency due to the inter-stage coupling capacitor (C_c) can be calculated considering the Z_{o1} and Z_{i2} as,

$$f_{L(C_c)} = \frac{1}{2\pi(Z_{o1} + Z_{i2})C_c} = \frac{1}{2\pi(r_{o1} \parallel R_{C1} + R_{B2} \parallel h_{ie2})C_c}$$

or,
$$f_{L(C_c)} = \frac{1}{2\pi(100 \text{ k} \parallel 5.1 \text{ k} + 5.54 \text{ k} \parallel 1.3 \text{ k})1 \text{ } \mu\text{F}} \approx 26 \text{ Hz}$$

To calculate the lower cut-off frequencies, we will use Equ.(11-52) of Chapter 11,

Equ.(11-52),
$$f_{L1(C_E)} = \frac{1}{2\pi[(r_s \parallel R_B + h_{ie})/h_{fe}][R_E]C_E}$$

For 1st stage,
$$f_{L1(C_E)} = \frac{1}{2\pi[(0.6 \text{ k} \parallel 5.54 \text{ k} + 1.3 \text{ k})/100][1.5 \text{ k}] \times 47 \text{ } \mu\text{F}} \approx 186 \text{ Hz [Ans.]}$$

For 2nd stage,
$$f_{L2(C_E)} = \frac{1}{2\pi[(Z_{o1} \parallel R_B + h_{ie})/h_{fe}][R_E]C_E}$$

or,
$$f_{L2(C_E)} = \frac{1}{2\pi[(4.85 \text{ k} \parallel 5.54 \text{ k} + 1.3 \text{ k})/100][1.5 \text{ k}] \times 47 \text{ } \mu\text{F}} \approx 89 \text{ Hz [Ans.]}$$

Therefore, the overall lower cut-off frequency of the two-stage amplifier will be little larger than 186 Hz [Ans.].

Comments: If the two lower cut-off frequencies of two stages were equal, the exact lower cut-off frequency of the two-stage amplifier could have been calculated using Equ.(12-8).

12.7 Direct Coupling Method

As the name implies, **direct coupling** is the coupling method in which the output of one stage is directly connected to the input of the next stage. Therefore, both the DC and AC voltages of the output of one stage will be identical to those of the input of next stage. The main advantages of direct coupling are: it requires less components and it does not affect the low frequency response of the individual amplifier stages. Direct coupling is commonly used in differential amplifiers and in operational amplifiers IC that will

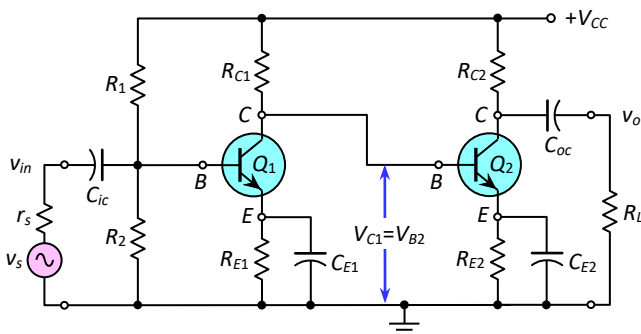


Fig.12-7: Two-stage direct coupled CE amplifier

be discussed later. Fig.12- 7 shows a direct coupled multistage amplifier consisting of two stages. Note that the DC voltage level of the output of first stage is same as the DC voltage level of the input of the second stage, i.e., $V_{C1} = V_{B2}$. In fact, the DC voltage level of the first stage is used to bias the base of the second stage. If we avoid using any capacitor in direct coupled amplifier, it can be used as AC amplifier as well as DC amplifier. In amplifier that uses single power supply, like Fig.12-7, it is not possible to connect the source and load directly without using coupling capacitors. Because they will change the DC bias conditions and the load will get a DC voltage level. To connect the source and load directly to the amplifier, we have to use split power supply as shown in Fig.12-8. If we bias the first stage in such a way that $V_{B1} = 0\text{ V}$, then we can connect the source directly to the input of the first stage.

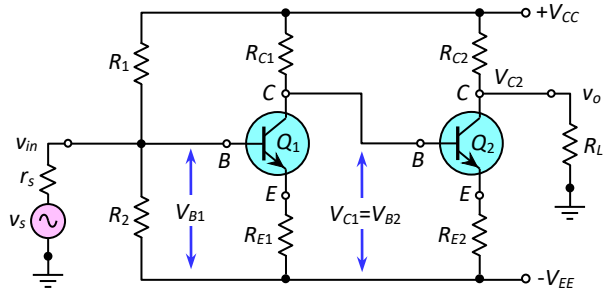


Fig.12-8: Two-stage direct coupled CE amplifier with split supply

The source resistance will not change the bias condition, similarly the base voltage will not affect the source, v_s . In the same way, if we bias the collector of last stage to zero voltage, that is $V_{C2} = 0$, then we can directly connect the load to this point. The load will not change the DC bias voltage level, and no DC current will flow through the load. Note that there is no coupling capacitor in this amplifier.

Direct coupled amplifiers suffer from *drift*, which is an unwanted change in output voltage without any change in supply voltages. The output also changes with the age or time, and change in β of transistor.

The main advantages of direct coupled amplifiers are: simple circuit arrangement and less expensive as they need minimum number of components, can be used to amplify DC and low frequency signals. The main disadvantages of these amplifiers are: they cannot amplify very high frequency signal, and they are less stable. These types of amplifiers with some other additional output stages are used in ICs.

12.7.1 AC Analysis of Direct Coupled Amplifiers

AC analysis of direct coupled multistage amplifier is similar to RC coupled amplifiers described above. The hybrid equivalent circuit of the two-stage amplifier of Fig.12-8 is shown in Fig.12-9. This equivalent circuit is same as the equivalent circuit of Fig.12-5, except, there is no R_{B2} in the second stage.

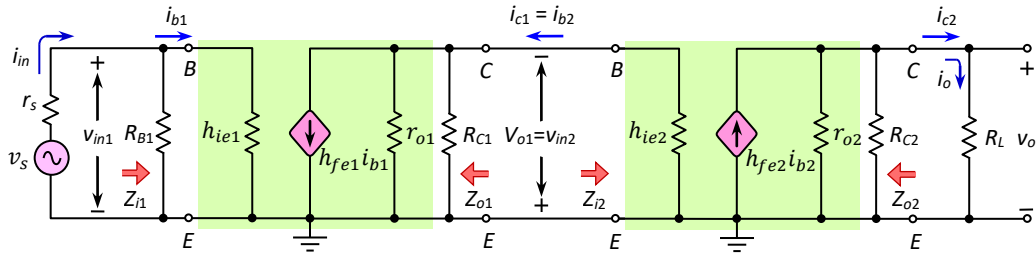


Fig.12-9: AC equivalent circuit of two-stage direct coupled CE amplifier

All the equations described for RC coupled amplifier will be applicable for this direct coupled amplifier, except Z_{i2} . From Fig.12-9, the input impedance of the second stage will be,

$$Z_{i2} = h_{ie2}$$

Therefore, the overall voltage gain of this multistage amplifier will be,

$$A_{vm} = \frac{h_{fe1}(R_{C1} \parallel h_{ie2})}{h_{ie1}} \times \frac{h_{fe2}(R_{C2} \parallel R_L)}{h_{ie2}} \quad (12-20)$$

12.7.2 Design of Direct Coupled Multistage Amplifiers

In designing the direct coupled multistage amplifier, first we have to design stage-1 using the method described in Chapter 10. In designing stage-2, we have to use the collector voltage of stage-1 ($V_{C1} = V_{CC} - I_C R_C$) as the base voltage of Q_2 . Here, V_{C1} will work as a supply voltage for the 2nd state with internal resistance of $Z_{o1} \approx R_{C1}$. Therefore, the base current of Q_2 can be calculated as,

$$i_{b2} = \frac{V_{C1} - V_{BE2}}{R_{C1} + (\beta + 1)R_{E2}} \quad (12-21)$$

Value of V_{C1} can be set, by estimating satisfactory voltage levels for V_{E1} , and V_{CE1} . As discussed before, the value of V_E can be 3 V to 5 V depending on biasing voltage and V_{CE} can be 1 V to 3 V. Once these values are set, V_{C1} will be

$$V_{C1} = V_{E1} + V_{CE1}$$

The base voltage, and emitter voltage of Q_2 will be,

$$V_{B2} = V_{C1} = V_{E1} + V_{CE1}$$

and,

$$V_{E2} = V_{B2} - V_{BE2} = V_{C1} - V_{BE2} = V_{E1} + V_{CE1} - V_{BE2}$$

The values of R_C should not be very larger or smaller than the load resistance connected to them. Therefore, R_{C1} should be comparable to $Z_{i2} = h_{ie2}$, and R_{C2} should be comparable to R_L .

Example 12-5

Design a direct coupled multistage amplifier with supply voltage of 12 V, and transistor having $h_{fe} = 100$, and $h_{ie} = 1 \text{ k}\Omega$. Assume $R_L = 4.7 \text{ k}\Omega$, $r_s = 600 \Omega$, and $f_L = 100 \text{ Hz}$.

Solution:

We will refer the multistage amplifier of Fig.12-8. Let us assume, $V_{E1} = 3 \text{ V}$, $V_{CE1} = V_{CE2} = 3 \text{ V}$, and $I_{C1} = I_{C2} = 1 \text{ mA}$. Therefore,

$$V_{C1} = V_{E1} + V_{CE1} = 3 \text{ V} + 3 \text{ V} = 6 \text{ V}$$

$$\therefore V_{B2} = V_{C1} = 6 \text{ V}$$

$$\therefore V_{E2} = V_{B2} - V_{BE2} = 6 \text{ V} - 0.7 \text{ V} = 5.3 \text{ V}$$

$$R_{E2} = \frac{V_{E2}}{I_{E2}} \approx \frac{V_{E2}}{I_{C2}} = \frac{5.3 \text{ V}}{1 \text{ mA}} = 5.3 \text{ k} \text{ (use standard 5.1 k)}$$

$$\therefore V_{RC2} = V_{CC} - V_{E2} - V_{CE2} = 12 \text{ V} - 5.3 \text{ V} - 3 \text{ V} = 3.7 \text{ V}$$

$$\therefore R_{C2} = \frac{V_{RC2}}{I_{C2}} = \frac{3.7 \text{ V}}{1 \text{ mA}} = 3.7 \text{ k} \text{ (use standard 3.6 k)}$$

Therefore, the exact value of V_{CE2} will be,

$$V_{CE2} = V_{CC} - I_{C2}(R_{C2} + R_{E2}) = 12 \text{ V} - 1 \text{ mA}(3.6 \text{ k} + 5.1 \text{ k}) = 3.3 \text{ V}$$

For the first transistor,

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{V_{CC} - V_{C1}}{I_{C1}} = \frac{12 \text{ V} - 6 \text{ V}}{1 \text{ mA}} = 6 \text{ k} \text{ (use standard 5.6 k)}$$

$$R_{E1} = \frac{V_{E1}}{I_{E1}} \approx \frac{V_{E1}}{I_{C1}} = \frac{3 \text{ V}}{1 \text{ mA}} = 3 \text{ k}$$

$$I_{B1} = I_{B2} = \frac{I_C}{h_{fe}} = \frac{1 \text{ mA}}{100} = 10 \mu\text{A}$$

Let $I_{R2} = 10 \times I_{B1} = 100 \mu\text{A}$

$$V_{B1} = V_{E1} + V_{BE1} = 3 \text{ V} + 0.7 \text{ V} = 3.7 \text{ V}$$

$$\therefore R_2 = \frac{V_{R2}}{I_{R2}} = \frac{V_{B1}}{I_{R2}} = \frac{3.7 \text{ V}}{100 \mu\text{A}} = 37 \text{ k} \text{ (use standard 3.6 k)}$$

$$I_{R1} = I_{R2} + I_{B1} = 100 \mu\text{A} + 10 \mu\text{A} = 110 \mu\text{A}$$

$$\therefore R_1 = \frac{V_{R1}}{I_{R1}} = \frac{V_{CC} - V_{R2}}{I_{R1}} = \frac{12 \text{ V} - 3.7 \text{ V}}{110 \mu\text{A}} \approx 75 \text{ k}$$

Let's check Equ.(12-9),

$$i_{b2} = \frac{V_{C1} - V_{BE2}}{R_{C1} + (\beta + 1)R_{E2}} = \frac{6 \text{ V} - 0.7 \text{ V}}{5.6 \text{ k} + (100 + 1)5.1 \text{ k}} = 10 \mu\text{A}$$

That is Equ.(12-9) gives the same base current as we have calculated before. If this value were less than I_{C2}/h_{fe2} , we have to decrease the value of R_{C1} .

$$\text{Using Equ.(10-8), } C_{ic} = \frac{10}{2\pi f_1(r_s + h_{ie1})} = \frac{10}{2\pi \times 100 \times (0.6 \text{ k} + 1 \text{ k})} \approx 10 \mu\text{F}$$

Using Equ.(10-10),
$$C_{oc} = \frac{10}{2\pi f_L(R_{C2} + R_L)} = \frac{10}{2\pi \times 100(3.6 \text{ k} + 4.7 \text{ k})} \approx 2 \mu\text{F}$$

Using Equ.(12-19),
$$C_{E1} = C_{E2} = \frac{1 + h_{fe}}{2\pi[0.65 f_{L(m)}]h_{ie}} = \frac{1 + 100}{2\pi[0.65 \times 100]1\text{k}} \approx 247 \mu\text{F}$$

Comments: We could have used lower value of R_{C1} than 5.6 k. Because 5.6 k is much larger than $Z_{i2} = h_{ie} = 1 \text{ k}$.

12.7.3 Direct Coupled Amplifier with Darlington Pair

It is very common to use a CC amplifier at the last stage of a multistage amplifier to amplify the current. High current transistors are necessary for this purpose. But the β value of high current transistors is low. That's why sometimes Darlington pair is employed. Fig.12-10 shows a two-stage amplifier. The first stage is CE amplifier and the second stage is a CC amplifier with Darlington pair.

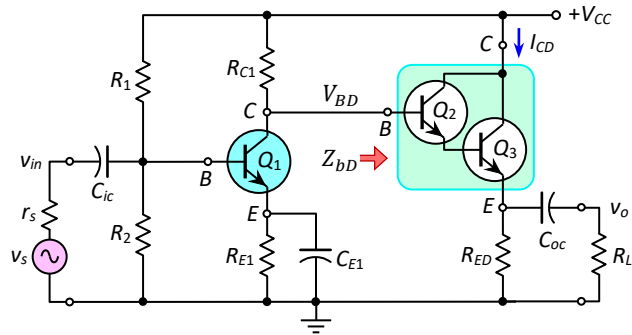


Fig.12-10: Two-stage direct coupled amplifier consisting of a CE amplifier and a CC amplifier with Darlington pair

According to Equ.(5-128) described in Chapter 5,

$$\beta_D = (\beta_2\beta_3 + \beta_2 + \beta_3) \quad (12-22)$$

and the AC resistance of Darlington transistor is (described in Chapter 5),

$$r_{eD} = 2r_e = 2 \times (26 \text{ mV}/I_{CD}) \quad (12-23)$$

The resistance seen from the base of Darlington pair will be,

$$Z_{bD} \approx \beta_D(r_{eD} + R_{ED} \parallel R_L) \quad (12-24)$$

Voltage gain of the first stage will be,

From Equ.(6-10),
$$A_{v1} \approx -\frac{(R_{C1} \parallel Z_{bD})}{r_{e1}} \quad (12-25)$$

Voltage gain of the second stage will be,

From Equ.(6-37),
$$A_{v2} = \frac{R_{ED} \parallel R_L}{r_{eD} + R_{ED} \parallel R_L} \quad (12-26)$$

The overall gain of this amplifier will be,

$$A_v = \frac{v_o}{v_{in}} \approx A_{v1}A_{v2} = -\frac{(R_{C1} \parallel Z_{bD})}{r_{e1}} \times \frac{R_{ED} \parallel R_L}{r_{eD} + R_{ED} \parallel R_L} \quad (12-27)$$

We can also find this value using hybrid parameters as well.

Example 12-6

The transistors used in the multistage amplifier of Fig.12-10, have $\beta_1 = 150$, $\beta_2 = 150$, and $\beta_3 = 100$. Assuming $V_{CC} = 12\text{ V}$, $R_1 = 30\text{ k}$, $R_2 = 6.8\text{ k}$, $R_{C1} = R_{C2} = 5.1\text{ k}$, $R_E = 1.5\text{ k}$, and $R_L = 10\text{ }\Omega$ calculate A_v , A_i , Z_i , and Z_o of individual stages and the multistage amplifier. Consider, $C_{ic1} = 5.6\text{ }\mu\text{F}$, $C_{oc2} = 1\text{ }\mu\text{F}$, and $C_E = 47\text{ }\mu\text{F}$.

Solution:

Using Equ.(12-22), $\beta_D = (\beta_2\beta_3 + \beta_2 + \beta_3) = (150 \times 100 + 150 + 100) = 15250$

We will use the exact analysis of CE amplifier used in the first stage. Therefore,

$$E_{Th} = \frac{V_{CC}R_2}{R_1 + R_2} = \frac{12\text{ V} \times 7.5\text{ k}}{39\text{ k} + 7.5\text{ k}} \approx 1.94\text{ V}$$

$$R_{Th} = \frac{R_1R_2}{R_1 + R_2} = \frac{39\text{ k} \times 7.5\text{ k}}{39\text{ k} + 7.5\text{ k}} \approx 6.3\text{ k} = R_{B1}$$

Base current of Q_1 will be

$$I_{B1} = \frac{E_{Th} - V_{BE1}}{R_{Th} + (\beta_1 + 1)R_{E1}} = \frac{1.94\text{ V} - 0.7\text{ V}}{6.3\text{ k} + (200 + 1)1\text{ k}} \approx 6\text{ }\mu\text{A} = I_{B2}$$

$$\therefore I_{C1} = \beta_1 I_{B1} = 200 \times 6\text{ }\mu\text{A} = 1.2\text{ mA}$$

$$\therefore V_{CE1} = V_{CC} - I_{C1}(R_{C1} + R_{E1}) = 12\text{ V} - 1.2\text{ mA}(4.7\text{ k} + 1\text{ k}) \approx 4.96\text{ V}$$

$$\therefore V_{C1} = V_{CC} - I_{C1}R_{C1} = 12\text{ V} - 1.2\text{ mA} \times 4.7\text{ k} = 6.36\text{ V}$$

The base voltage of the Darlington pair will be,

$$\therefore V_{BD} = V_{C1} = 6.36\text{ V}$$

Therefore, the emitter voltage will be 1.4 V lower than V_{BD} .

$$\therefore V_{ED} = V_{BD} - 1.4\text{ V} = 6.36\text{ V} - 1.4\text{ V} = 4.96\text{ V}$$

$$\therefore I_{ED} = \frac{V_{ED}}{R_{ED}} = \frac{4.96\text{ V}}{1\text{ mA}} = 4.96\text{ mA}$$

AC resistance of Darlington transistor is,

$$\text{Using Equ.(12-23), } r_{eD} = 2r_e = 2(26\text{ mV}/I_C) \approx 2(26\text{ mV}/I_E)$$

$$\therefore r_{eD} = 2(26\text{ mV}/4.96\text{ mA}) \approx 10.48\text{ }\Omega$$

$$\text{Using Equ.(12-24), } Z_{bD} \approx \beta_D(r_{eD} + R_E \parallel R_L)$$

$$\therefore Z_{bD} = 15250 \times (10.48\text{ }\Omega + 1000 \parallel 10)\text{ }\Omega = 310.81\text{ k}$$

$$\text{For } Q_1, r_{e1} = r_{e1} = 26\text{ mV}/I_{C1} \approx 26\text{ mV}/1.2\text{ mA} = 21.67\text{ }\Omega$$

Voltage gain of the first stage will be,

Using Equ.(12-25), $A_{v1} \approx -\frac{(R_{C1} \parallel Z_{bD})}{r_{e1}} = -\frac{(4.7 \text{ k} \parallel 310.81 \text{ k})}{21.67 \Omega} \approx -214 \text{ [Ans.]}$

Using Equ.(12-26), $A_{v2} = \frac{R_{ED} \parallel R_L}{r_{eD} + R_{ED} \parallel R_L} = \frac{1 \text{ k} \parallel 10 \Omega}{10.48 \Omega + 1 \text{ k} \parallel 10 \Omega} \approx 0.49 \text{ [Ans.]}$

Therefore, overall gain of this multistage amplifier is

$$A_{v(m)} = A_{v1} \times A_{v2} = -214 \times 0.49 \approx -104 \text{ [Ans.]}$$

Comments: The value of R_L is very low. If we connect such low R_L directly to output of a CE amplifier the voltage gain will drastically fall down. Here, the CC amplifier has isolated R_L from the CE amplifier.



12.8 Transformer Coupled Amplifier

The third coupling method is the **transformer coupling**. Transformer can couple AC signal from one stage to another while isolating the DC voltage levels. As shown in Fig.12-11, the primary winding of a transformer is connected at the output of first stage and the secondary of that transform is connected to the input of the next stage. The output voltage of the first stage will be developed across the primary coil and the transformer will couple this AC voltage to the

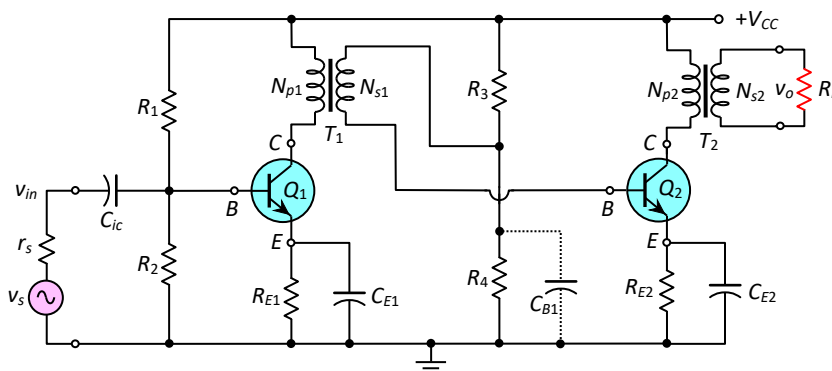


Fig.12-11: Two-stage transformer coupled CE amplifier

secondary coil using mutual induction process. The voltage induced in the secondary coil will be applied to the input of the next stage. Sometimes, we use a capacitor (C_{B2}) across the bias resistor (R_4) of second stage. In that case the transferred AC voltage will fully be applied to the base of Q_2 . The main advantages of transformer coupling are: low DC power dissipation and the capability of impedance matching by changing the turn ratio. Impedance matching ensures maximum power transfer from one stage to another. The main disadvantages of transformer coupling are: transformers are bulky and costly, and their poor frequency response. The reactance of the transformer coils, and the winding capacitance tend to reduce the useable bandwidth of these amplifiers. However, the transformer coupled amplifiers are used in narrowband applications, such as RF (radio frequency) amplifiers.

In designing the biasing circuit, we have to consider the primary DC resistance (R_p) of the transformer. As R_p is very small, sometimes we neglect this value and divide V_{CC} equally as $V_E = V_{CE} = V_{CC}/2$. Calculations of other biasing components are same as before. The bias collector current of this amplifier should not be very large that can saturate the core of the transformer.

Transformer coupled amplifiers are mainly used for *impedance matching*, for *power amplification*, and for *maximum power transfer*.

12.8.1 AC Analysis of Transformer Coupled Amplifier

To draw the equivalent circuit of this amplifier, we must calculate the AC resistance seen by the collectors of the transistors. Using the impedance feedback principle of transformer we can write,

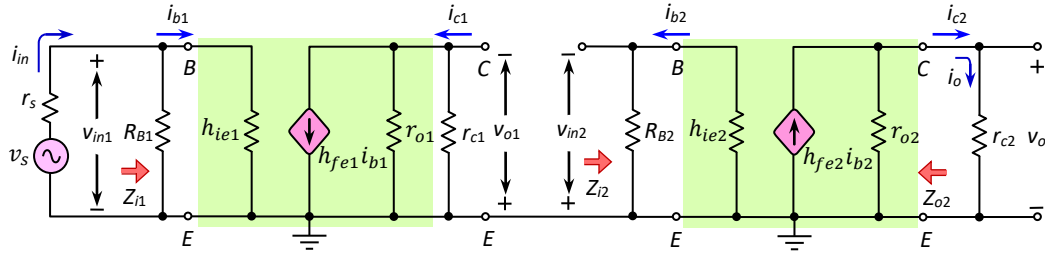


Fig.12-12: AC equivalent circuit of two-stage transformer coupled CE amplifier

$$r_{c1} = \left(\frac{N_{p1}}{N_{s1}} \right)^2 R_{B2} \parallel h_{ie2} \quad (12-28)$$

where, N_p , and N_s are number of turns in primary and secondary coils, respectively. Similarly,

$$r_{c2} = \left(\frac{N_{p2}}{N_{s2}} \right)^2 R_L \quad (12-29)$$

Using these AC resistances, the equivalent circuit of the two-stage amplifier of Fig.12-11 is drawn in Fig.12-12. Look at the equivalent circuit, the transformer provides isolation between the stages.

The input and output impedance of this amplifier will be,

$$Z_i = Z_{i1} = R_{B1} \parallel h_{ie1} \quad (12-30)$$

and,

$$Z_o = Z_{o2} = r_{o2} \quad (12-31)$$

Input impedance of the second stage will be,

$$Z_{i2} = R_{B2} \parallel h_{ie2} \quad (12-32)$$

Voltage gain of the first stage will be,

$$A_{v1} = - \frac{h_{fe1}(r_{o1} \parallel r_{c1})}{h_{ie1}} \quad (12-33)$$

If r_{o1} is very large,

$$A_{v1} \approx - \frac{h_{fe1} r_{c1}}{h_{ie1}} \quad (12-34)$$

Voltage gain of the second stage will be,

$$A_{v2} = -\frac{h_{fe2}(r_{o2} \parallel r_{c2})}{h_{ie2}} \quad (12-35)$$

If r_{o2} is very large,

$$A_{v2} \approx -\frac{h_{fe2}r_{c2}}{h_{ie2}} \quad (12-36)$$

However, the output of the first stage (v_{o1}) will be step down by T_1 . So the input voltage of second stage will be,

$$v_{in2} = \left(\frac{N_{s1}}{N_{p1}}\right) v_{o1} \quad (12-37)$$

Similarly, the output of second stage will be step down by T_2 . So the voltage across the load will be,

$$v_L = \left(\frac{N_{s2}}{N_{p2}}\right) v_{o2} \quad (12-38)$$

Therefore, the overall voltage gain of the multistage amplifier will be the product of the individual voltage gain and transformer turn ratios. That is,

$$A_v = \left(\frac{N_{s1}}{N_{p1}}\right) \left(\frac{N_{s2}}{N_{p2}}\right) A_{v1} \times A_{v2} \quad (12-39)$$

Again, if we consider the effect of source resistance, the overall voltage gain will be,

$$A_v = \frac{v_o}{v_s} = \frac{Z_{i1}}{r_s + Z_{i1}} \left(\frac{N_{s1}}{N_{p1}}\right) \left(\frac{N_{s2}}{N_{p2}}\right) A_{v1} \times A_{v2} \quad (12-40)$$

Example 12-7

Assume that the transformer coupled amplifier of Fig.12-11, has been constructed using two transistors of $h_{fe2} = 100$, $r_0 = 100 \text{ k}$, and $V_{CC} = 12 \text{ V}$. Values of other components are,

$R_1 = R_3 = 30 \text{ k}$, $R_2 = R_4 = 6.8 \text{ k}$, $R_{E1} = R_{E2} = 1.5 \text{ k}$, $R_{p1} = R_{p2} = 20 \Omega$, $N_{p1}/N_{s1} = 2$, $N_{p2}/N_{s2} = 3$, $r_s = 600 \Omega$, and $R_L = 1 \text{ k}$.

Calculate the DC operating points of the transistors, voltage gain of individual stages, and overall voltage gain v_L/v_s of the two-stage amplifier.

Solution:

DC biasing components of the transistors are symmetrical. Therefore,

$$E_{Th} = \frac{V_{CC}R_2}{R_1 + R_2} = \frac{12 \text{ V} \times 6.8 \text{ k}}{30 \text{ k} + 6.8 \text{ k}} \approx 2.21 \text{ V}$$

$$R_{Th} = \frac{R_1R_2}{R_1 + R_2} = \frac{30 \text{ k} \times 6.8 \text{ k}}{30 \text{ k} + 6.8 \text{ k}} \approx 5.54 \text{ k} = R_B$$

Base current of Q_1 , and Q_2 will be

$$I_{B1} = \frac{E_{Th} - V_{BE1}}{R_{Th} + (\beta + 1)R_{E1}} = \frac{2.21 \text{ V} - 0.7 \text{ V}}{5.54 \text{ k} + (100 + 1)1.5 \text{ k}} \approx 9.62 \mu\text{A} = I_{B2}$$

$$\therefore I_{C1} = \beta I_{B1} = 100 \times 9.62 \mu\text{A} = 962 \mu\text{A} = I_{C2}$$

$$\therefore V_{CE1} = V_{CE2} = V_{CC} - I_C(R_p + R_E) = 12 \text{ V} - 962 \mu\text{A}(20 \Omega + 6.2 \text{ k}) \approx 6 \text{ V}$$

DC operating points of the transistors will be, $I_C = 962 \mu\text{A}$, and $V_{CE} = 6 \text{ V}$ [Ans.]

$$\text{Now, } r_{e1} = r_{e2} = \frac{26 \text{ mV}}{I_E} \approx \frac{26 \text{ mV}}{I_C} = \frac{26 \text{ mV}}{962 \mu\text{A}} \approx 27 \Omega$$

$$h_{ie1} = h_{ie2} = \beta r_e \approx 100 \times 27 \Omega = 2.7 \text{ k}$$

$$\text{Using Equ.(12-28), } r_{c1} = \left(\frac{N_{p1}}{N_{s1}}\right)^2 R_{B2} \parallel h_{ie2} = 2^2 \times 5.54 \text{ k} \parallel 2.7 \text{ k} \approx 7.26 \text{ k}$$

$$\text{Using Equ.(12-29), } r_{c2} = \left(\frac{N_{p2}}{N_{s2}}\right)^2 R_L = 3^2 \times 1 \text{ k} = 9 \text{ k}$$

Voltage gain of the first stage,

$$\text{Using Equ.(12-31), } A_{v1} = -\frac{h_{fe1}(r_{o1} \parallel r_{c1})}{h_{ie1}} = -\frac{100(100 \text{ k} \parallel 7.26 \text{ k})}{2.7 \text{ k}} \approx -251 \text{ [Ans.]}$$

Voltage gain of the second stage,

$$\text{Using Equ.(12-32), } A_{v2} = -\frac{h_{fe1}(r_{o1} \parallel r_{c1})}{h_{ie1}} = -\frac{100(100 \text{ k} \parallel 9 \text{ k})}{2.7 \text{ k}} \approx -306 \text{ [Ans.]}$$

$$\text{Input impedance is, } Z_{i1} = R_{B1} \parallel h_{ie1} = 5.54 \text{ k} \parallel 2.7 \text{ k} \approx 1.82 \text{ k}$$

$$\text{Using Equ.(12-38), } A_v = \frac{v_o}{v_s} = \frac{Z_{i1}}{r_s + Z_{i1}} \left(\frac{N_{s1}}{N_{p1}}\right) \left(\frac{N_{s2}}{N_{p2}}\right) A_{v1} \times A_{v2}$$

$$\text{or, } A_v = \frac{1.82 \text{ k}}{0.6 \text{ k} + 1.82 \text{ k}} \left(\frac{1}{2}\right) \left(\frac{1}{3}\right) (-251)(-306) \approx 9627 \text{ [Ans.]}$$

Comments: By changing the turn ratios of the transformer, we can change the gain of this amplifier and can connect R_L is very small value without reducing the gain.

12.8.2 Frequency Response of Transformer Coupled Amplifier

The frequency response of transformer coupled amplifier is not so good as that of RC coupled and direct coupled amplifiers. The frequency response of a transformer coupled amplifier is shown in Fig.12-13. From this figure, we find that the gain is flat only for a small portion of frequency range. The output voltage is equal to the collector

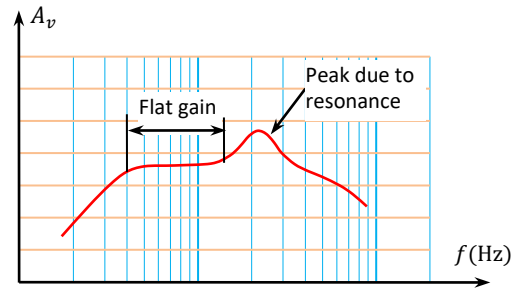


Fig.12-13: Typical frequency response of transformer coupled amplifier

current multiplied by the reactance of primary coil of the transformer. At low frequencies, the reactance of the primary coil begins to fall, resulting in reduction of gain. After the flat-band frequencies, there is a region where the gain is higher than the flat-band region. This portion is due to the resonance of the transformer reactance with the parasitic capacitances of the transistors. After that peak, gain falls again with frequency. At this range, the capacitance between turns of winding of transformer becomes effective and act as bypass capacitor and tends to short the output voltage. Therefore, a transformer coupled amplifier can properly amplify a very small range of signals – like audio signals. For a wide range frequency signal – like music (includes frequencies from different instruments of different frequencies), this amplifier will produce both amplitude and phase distortions.

12.8.3 Design of Transformer Coupled Multistage CE Amplifiers

Except the transformer, the design process of this amplifier is same as discussed earlier. As there is no collector resistor, we have to consider the primary DC resistance (R_p) of the transformer. As R_p is very small, we assign larger voltage to V_E , and V_{CE} . Calculation process of other biasing components is same as before. The values of coupling capacitors can be calculated using the process described for RC coupled amplifier. If the bypass capacitor C_{B2} is used, its value has to be calculated as,

$$X_{CB2} = \frac{h_{ie2}}{10}$$

or,

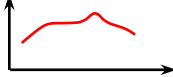
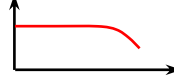
$$C_{B2} = \frac{10}{2\pi f_{L(m)} h_{ie}} \quad (12-41)$$

We have learnt that the main advantages of transformer coupled amplifier are- impedance matching, minimum DC power loss. For these advantages, transformer-coupling is mainly used for power amplifiers. The design process of transformer coupled power amplifier is described in Chapter 15 [H_C_atten].

12.9 Comparison of Different Coupling Methods

We have discussed three types of coupling methods for multistage amplifiers. Every method has its advantages and disadvantages. In the following table, the main features of three coupling methods are given.

Parameters	RC Coupling	Transformer Coupling	Direct Coupling
Coupling Element	Resistor-Capacitor	Transformer	None
Signal Coupled	Only AC	Only AC	Both AC & DC
DC Blocking	Present	Present	Absent
Drift	Absent	Absent	Present

Impedance matching	Not good	Excellent	Good
Required space & cost	Less	More	Least
Application	Public addressing system	Power amplifier	Integrated circuit
Frequency Response			

12.10 Cascode Amplifier

In BJT Biasing Chapter (Chapter 5), we have discussed DC analysis of cascode amplifier. In that circuit we have used a single voltage divider network to bias both the transistors. But we can also use separate voltage divider network to bias the transistors as shown in Fig.12-14. R_1, R_2 set the biasing voltage for Q_1 and R_3, R_4 do that for Q_2 . This circuit structure helps to understand the cascode amplifier as multistage arrangement. The output of the first CE amplifier (taken for the collector of Q_1) is directly applied to the input (emitter) of the next CB amplifier. The input impedance of the CB amplifier will work as the collector resistance of the CE amplifier. The base of Q_2 is AC short to ground by C_{B2} capacitor and the output is taken from its collector. Therefore, cascode amplifier is a direct coupled two-stage amplifier.

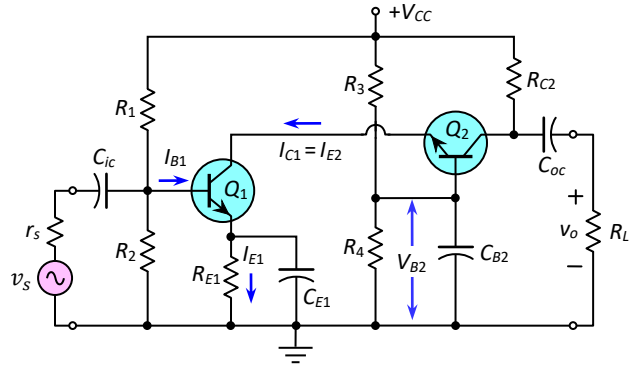


Fig.12-14: BJT cascode amplifier

As discussed earlier the exact base current can be calculated using Thevenin's equivalent circuit of the biasing network. Therefore,

$$E_{Th1} = \frac{V_{CC} \times R_2}{R_1 + R_2} \quad \text{and} \quad R_{Th1} = \frac{R_1 \times R_2}{R_1 + R_2}$$

The base current of Q_1 will be,

$$I_{B1} = \frac{E_{Th1} - V_{BE1}}{R_{Th1} + (\beta_1 + 1)R_{E1}}$$

$\therefore$

$$I_{C1} = \beta_1 I_{B1} \approx I_{E1}$$

$\therefore$

$$V_{E1} = I_{E1} R_{E1}$$

$$V_{B2} \approx \frac{V_{CC} R_4}{R_3 + R_4}$$

$$\therefore V_{C1} = V_{B2} - V_{BE2} = V_{E2}$$

$$\therefore V_{CE1} = V_{C1} - V_{E1}$$

$$V_{E2} = V_{B2} - V_{BE2} = V_{C1}$$

$$\therefore I_{C2} \approx I_{E2} = I_{C1}$$

$$\therefore V_{C2} = V_{CC} - I_{C2} R_{C2}$$

$$\therefore V_{CE2} = V_{C2} - V_{E2}$$

The first stage produces 180° phase shift and the second stage produces 0° phase shift. Therefore, the overall phase shift of this amplifier is 180° . Cascode amplifier provides better frequency response, as the CB amplifier does not suffer from Miller's Effect. To reduce the Miller's effect in the CE amplifier, its gain is set very low. The main purpose of this CE amplifier is to provide better input impedance and current gain (which are larger than those of CB amplifier).

12.10.1 AC Analysis of Cascode Amplifier

For the mid-band frequency range, all the coupling capacitors will be short, and all the parasitic capacitances will be open. Moreover, for AC response the DC supply source has to be considered short. Thus, the small signal AC equivalent circuit of cascode amplifier is shown in Fig.12-15.

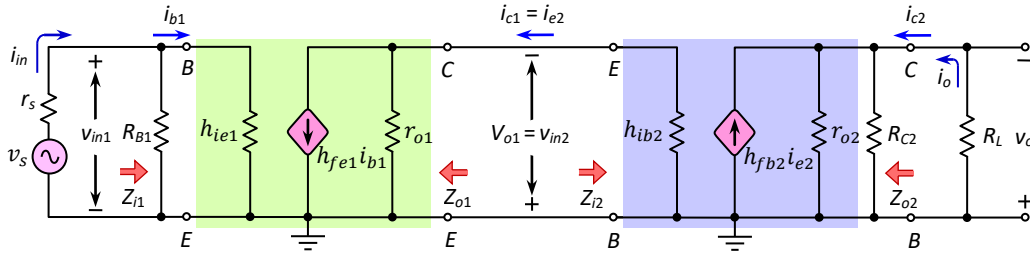


Fig.12-15: AC equivalent circuit of BJT cascode amplifier

If h_{ie1} and h_{ib2} are not known, their values can be calculated as,

$$h_{ie1} = h_{fe1} r_{e1} = h_{fe1} \frac{26 \text{ mV}}{I_{E1}}$$

and,

$$r_{e2} = \frac{26 \text{ mV}}{I_{E2}}$$

Voltage gain of the first stage will be,

$$A_{v1} = - \frac{h_{fe1} (r_{o1} \parallel h_{ib2})}{h_{ie1}} \quad (12-42)$$

$\therefore r_o$ is very large,

$$A_{v1} \approx - \frac{h_{fe1} h_{ib2}}{h_{ie1}} \quad (12-43)$$

Voltage gain of the second stage will be,

$$A_{v2} = \frac{h_{fb2}(r_{o2} \parallel R_{C2} \parallel R_L)}{h_{ib2}} \quad (12-44)$$

$\because r_o$ is very large,

$$A_{v2} \approx \frac{h_{fb2}R_{C2} \parallel R_L}{h_{ib2}} \quad (12-45)$$

The overall voltage gain of this multistage amplifier will be the product of the individual voltage gain. Therefore,

$$A_{v(m)} = A_{v1} \times A_{v2} \approx -\frac{h_{fe1}R_{C1} \parallel h_{ib2}}{h_{ie1}} \times \frac{h_{fb2}R_{C2} \parallel R_L}{h_{ib2}} \quad (12-46)$$

Example 12-8

The transistors used in the cascode amplifier of Fig.12-16 have $h_{fe} = 100$, $r_o = 100 \text{ k}$, $C_{be} = 10 \text{ pF}$, $C_{bc} = 5 \text{ pF}$, and $C_{ce} = 2 \text{ pF}$. Calculate the DC operating points of the transistors, voltage gain of individual stages, overall voltage gain, and higher cut-off frequency of the cascade amplifier.

Solution:

$$E_{Th1} = \frac{V_{CC}R_2}{R_1 + R_2} = \frac{12 \text{ V} \times 10 \text{ k}}{33 \text{ k} + 10 \text{ k}} \approx 2.79 \text{ V}$$

$$R_{Th1} = \frac{R_1R_2}{R_1 + R_2} = \frac{33 \text{ k} \times 10 \text{ k}}{33 \text{ k} + 10 \text{ k}} \approx 6.67 \text{ k} = R_{B1}$$

$$I_{B1} = \frac{E_{Th1} - V_{BE1}}{R_{Th1} + (\beta + 1)R_{E1}} = \frac{2.79 \text{ V} - 0.7 \text{ V}}{6.67 \text{ k} + (100 + 1)2.2 \text{ k}} \approx 9.132 \mu\text{A}$$

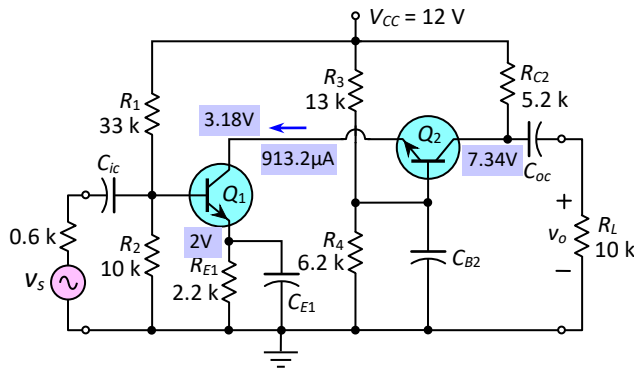


Fig.12-16: Cascode amplifier designed in Example 12-8

$$\therefore I_{C1} = \beta_1 I_{B1} = 100 \times 9.132 \mu\text{A} = 913.2 \mu\text{A} \approx I_{E1}$$

$$\therefore V_{E1} = I_{E1}R_{E1} = 913.2 \mu\text{A} \times 2.2 \text{ k} = 2 \text{ V}$$

$$V_{B2} \approx \frac{V_{CC}R_4}{R_3 + R_4} = \frac{12 \text{ V} \times 6.2 \text{ k}}{13 \text{ k} + 6.2 \text{ k}} \approx 3.88 \text{ V}$$

$$V_{C1} = V_{B2} - V_{BE2} = 3.88 \text{ V} - 0.7 \text{ V} = 3.18 \text{ V} = V_{E2}$$

$$\therefore V_{CE1} = 3.18 \text{ V} - 2 \text{ V} = 1.18 \text{ V}$$

$$\therefore I_{C2} \approx I_{E2} = I_{C1} = 913.2 \mu\text{A}$$

$$\therefore V_{C2} = V_{CC} - I_{C2}R_{C2} = 12 \text{ V} - 913.2 \mu\text{A} \times 5.1 \text{ k} = 7.34 \text{ V}$$

$$\therefore V_{CE2} = V_{C2} - V_{E2} = 7.34 \text{ V} - 3.18 \text{ V} = 4.16 \text{ V}$$

Therefore, the operating points of the transistors are,

$$Q_1 : I_{C1} = 913.2 \mu\text{A}, \quad V_{CE1} = 1.18 \text{ V [Ans.]}$$

$$Q_2 : I_{C2} = 913.2 \mu\text{A}, \quad V_{CE2} = 4.16 \text{ V [Ans.]}$$

$$r_{e1} = \frac{26 \text{ mV}}{I_{E1}} \approx \frac{26 \text{ mV}}{I_{C1}} = \frac{26 \text{ mV}}{913.2 \mu\text{A}} \approx 28.47 \Omega$$

$$h_{ie1} = h_{fe1}r_{e1} = 100 \times 28.47 \Omega \approx 2.85 \text{ k}$$

$$\therefore I_{C1} = I_{C2}, \quad \therefore r_{e2} = r_{e1} = 28.47 \Omega = h_{ib2}$$

As r_{o1} is very larger than h_{ib2} , we can neglect r_{o1} . Therefore,

$$\text{Using Equ.(12-43), } A_{v1} = -\frac{h_{fe1}h_{ib2}}{h_{ie1}} = -\frac{100 \times 28.47 \Omega}{2.85 \text{ k}} = -1$$

$$\text{Using Equ.(12-44), } A_{v2} = \frac{h_{fb2}(r_{o2} \parallel R_{C2} \parallel R_L)}{h_{ib2}} = \frac{1 \times (100 \text{ k} \parallel 5.1 \text{ k} \parallel 10 \text{ k})}{28.47 \Omega} \approx 115$$

Input impedance of the first stage is,

$$Z_{i1} = R_{B1} \parallel h_{ie1} = 6.67 \text{ k} \parallel 2.85 \text{ k} \approx 2 \text{ k}$$

Therefore, considering the effect of r_s , the overall voltage gain of the amplifier will be,

$$A_{v(m)} = \left(\frac{Z_{i1}}{r_s + Z_{i1}} \right) A_{v1} \times A_{v2} = \left(\frac{2 \text{ k}}{0.6 \text{ k} + 2 \text{ k}} \right) (-1)(115) \approx 88.5 [\text{Ans.}]$$

Higher cut-off frequency:

$$\text{Using Equ.(11-53), } C_{i1} = 10 \text{ pF} + 5 \text{ pF}(1 + 1) + 0 = 20 \text{ pF}$$

$$\text{Using Equ.(11-55), } f_{H(C_{i1})} = \frac{1}{2\pi \times 20 \text{ pF}(0.6 \text{ k} \parallel 2 \text{ k})} \approx 17.24 \text{ MHz [Ans.]}$$

As there is no Miller's effect in CB amplifier, $f_{H(C_{i1})}$ will be the lowest higher cut-off frequency in the cascade amplifier (unless limited by f_β). Therefore, the over-all higher cut-off frequency of the cascade amplifier will be a little smaller than $f_{H(C_{i1})}$.

Comments: In Example 11-9 [H_Ex_atten], the higher cut-off frequency was only 1.55 MHz. Though the parasitic capacitances have been considered the same in this cascode amplifier, the higher cut-off frequency has been increase to 17.24 MHz.

12.11 Multistage FET Amplifier

Already we have learnt that the gain of FET amplifier is comparatively low. So to get sufficient gain several FET amplifiers are sometimes cascaded and used as multistage amplifier. All of the three coupling methods can be used for FET multistage amplifiers. The properties of coupling methods remain same in FET amplifiers.

12.12 RC Coupled Multistage FET Amplifier

Fig.12-17 shows a two-stage RC coupled CS (common source) amplifier. Both stages are designed using voltage divider bias. The output of first stage is fed to the input of second stage by a coupling capacitor C_{oc} . DC analysis of individual stages can be done as discussed in Chapter 8 [H_C_atten]. For overall AC analysis, the AC equivalent circuit of this amplifier is shown in Fig.12-87.

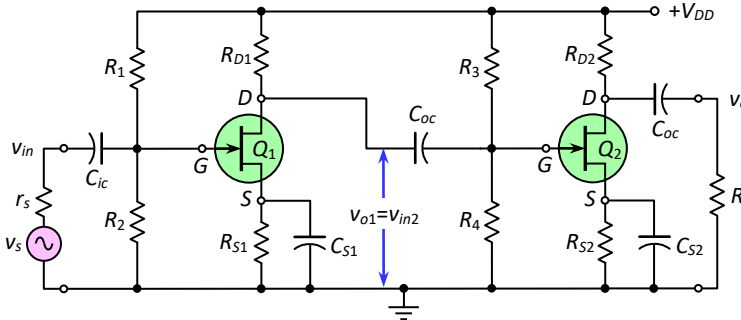


Fig.12-17: Two-stage RC coupled CS amplifier

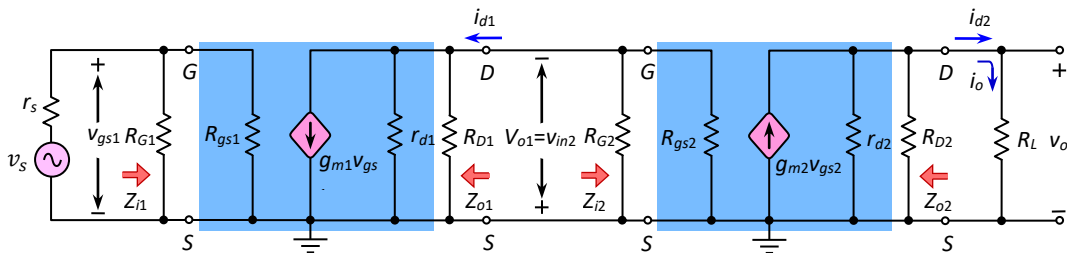


Fig.12-18: AC equivalent circuit of two-stage RC coupled CS amplifier

The input and output impedance of this amplifier will be,

$$Z_i = Z_{i1} = R_{G1} || R_{gs1} \approx R_{G1} \quad (12-47)$$

and,

$$Z_o = Z_{o2} = r_{d2} || R_{D2} \approx R_{D2} \quad (12-48)$$

Input impedance of the second stage will be,

$$Z_{i2} = R_{G2} \parallel R_{gs2} \approx R_{G2} \quad (12-49)$$

Voltage gain of the first stage will be,

$$A_{v1} = -g_{m1}(r_{d1} \parallel R_{D1} \parallel R_{G2} \parallel R_{gs2}) \quad (12-50)$$

$\because R_{gs1}$ and r_{d1} is very large,

$$A_{v1} = -g_{m1}(R_{D1} \parallel R_{G2}) \quad (12-51)$$

Voltage gain of the second stage will be,

$$A_{v2} = -g_{m2}(r_{d2} \parallel R_{D2} \parallel R_L) \quad (12-52)$$

$\because r_{d2}$ is very large,

$$A_{v2} = -g_{m2}(R_{D2} \parallel R_L) \quad (12-53)$$

The overall voltage gain of the multistage amplifier will be the product of the individual voltage gain. Therefore,

$$A_v \approx A_{v1} \times A_{v2} = g_{m1}(R_{D1} \parallel R_{G2}) \times g_{m2}(R_{D2} \parallel R_L) \quad (12-54)$$

Using the design concept of single stage CS amplifier this multistage circuit can also be designed. Only we have to modify the equation to calculate the value of C_s as,

$$C_s = \frac{g_m}{2\pi \cdot 0.65 f_{L(m)}} \quad (12-55)$$

Example 12-9

Determine the position of the operating points for the voltage divider bias circuit of Fig.12-17. The JFETs used in the multistage amplifier of Fig.12-17 has $V_p = -3$ V, and $I_{DSS} = 8$ mA. Determine the DC operating point and voltage gain of the individual stages of this amplifier. Also determine the overall voltage gain of the multistage amplifier. Assume $R_1 = R_3 = 1$ M, $R_2 = R_4 = 240$ k, $R_{D1} = R_{D2} = 2$ k, $R_{S1} = R_{S2} = 1$ k, $V_{DD} = 20$ V.

Solution:

Since, graphical method visualize the position of the operating points, here this method is used.

From $V_p = -4$ V and $I_{DSS} = 8$ mA

The value of V_G can be calculated using Equ.(8-18),

$$V_G = \frac{V_{DD} R_2}{R_1 + R_2} = \frac{20\text{V} \times 240\text{ k}}{240\text{ k} + 1000\text{ k}} = 3.87\text{ V}$$

From Equ.(8-18), the bias line of this circuit will be,

$$V_{GS} = V_G - I_D R_S = 3.87\text{ V} - (1\text{ k})I_D$$

Using Equ.(8-20),
$$I_D = I_{DSS} \left(1 - \frac{V_G - I_D R_S}{V_P} \right)^2 = 12 \text{ mA} \left(1 - \frac{3.87 - 1\text{k} \times I_D}{4 \text{ V}} \right)^2$$

Solving this equation we get $I_D = 4.78 \text{ mA}$ [Ans.]

Using Equ.(8-21), $V_{DS} = V_{DD} - I_D(R_D + R_S) = 20 \text{ V} - 4.78 \text{ mA}(2 \text{ k} + 1 \text{ k}) = 5.66 \text{ V}$ [Ans.]

The input and output impedance of this amplifier will be,

Using Equ.(12-47), $Z_i \approx R_{G1} = R_1 \parallel R_2 = 1 \text{ M} \parallel 240 \text{ k} \approx 193.54 \text{ k}$

and, using Equ.(12-48), $Z_o = Z_{o2} \approx R_{D2} = 2 \text{ k}$

Input impedance of the second stage will be,

Using Equ.(12-49), $Z_{i2} \approx R_{G2} = R_3 \parallel R_4 = 1 \text{ M} \parallel 240 \text{ k} \approx 193.54 \text{ k}$

Using Equ.(7-12),
$$g_{m1} = g_{m2} = \frac{2\sqrt{I_{DSS}I_D}}{|V_P|} = \frac{2\sqrt{8 \text{ mA} \times 4.78 \text{ mA}}}{|4 \text{ V}|} \approx 3.1 \text{ mS}$$

Voltage gain of the first stage will be,

Using Equ.(12-51), $A_{v1} = -g_{m1}(R_{D1} \parallel R_{G2}) = -3.1 \text{ mS}(2 \text{ k} \parallel 230.77 \text{ k})$

or, $A_{v1} \approx -6.15$ [Ans.]

Voltage gain of the second stage will be,

Using Equ.(12-53), $A_{v2} = -g_{m2}(R_{D2} \parallel R_L) = -3.1 \text{ mS}(2 \text{ k} \parallel 10 \text{ k})$

or, $A_{v2} \approx -5.17$ [Ans.]

The overall voltage gain of the multistage amplifier will be the product of the individual voltage gain. Therefore,

$$A_v \approx A_{v1} \times A_{v2} = (-6.15) \times (-5.17) \approx 31.8 \quad (12-56)$$

Comments: Gain of FET amplifier is comparatively lower than that of BJT amplifiers.

12.13 Direct Coupled CS Amplifier

Direct couple amplifier has some advantages over RC and transformer coupled amplifiers. Like BJT amplifiers, direct coupling method can be used for FET amplifiers. Fig.12-19 shows a two stage direct couple CS amplifier. Here, the first stage employs voltage divider bias. We can also use self bias. The drain voltage (V_{D1}) of Q_1 works as the gate voltage (V_{G2}) for Q_2 . Although, in this circuit we have used input and output coupling capacitors and source bypass capacitors for every stage, but these capacitors can be avoided, using proper DC voltage levels through biasing. In that case this amplifier will be able to amplify DC signals.

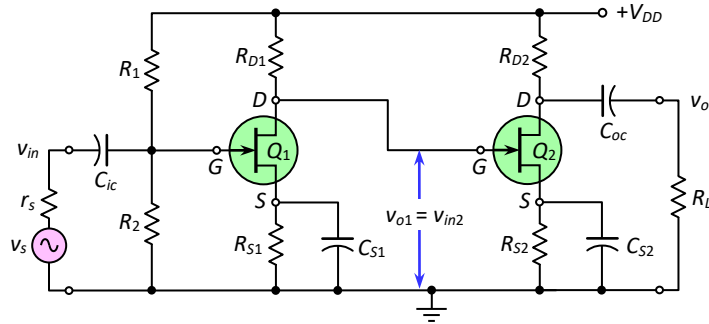


Fig.12-19: Two-stage RC coupled CS amplifier

Example 12-10

Suppose the two-stage direct coupled amplifier of Fig.12-19 uses supply voltage of $\pm 12\text{ V}$, $R_1 = R_2 = R_3 = R_4 = 1\text{ M}$, $R_{S1} = R_{S2} = 2.2\text{ k}$, $R_{D1} = 1.2\text{ k}$, $R_{D2} = 300\ \Omega$, and $R_L = 4.7\text{ k}$. Assume that the JFETs have $I_{DSS} = -3\text{ V}$, $I_{DSS} = 12\text{ mA}$. Determine the operating points and voltage gain of the individual stages and the overall gain of the two-stage amplifier.

Solution:

Using Equ.(8-32),
$$V_{G1} = \frac{R_2 V_{DD}}{R_1 + R_2} - \frac{R_1 V_{SS}}{R_1 + R_2} = 0\text{ V}$$

Using Equ.(8-33),
$$V_{GS1} = V_{G1} + |V_{SS}| - I_{D1} R_{S1} = 12\text{ V} - 2.2\text{ k} \times I_{D1}$$

Putting this value into Shockley's equation,

$$I_{D1} = I_{DSS} [1 - (12\text{ V} - 2.2\text{ k} \times I_{D1}) / (-3\text{ V})]^2$$

Solving this equation, we get $I_{D1} = 5.87\text{ mA}$ [Ans.]

Using Equ.(8-33),
$$V_{DS1} = 12\text{ V} + 12\text{ V} - 5.87\text{ mA}(2.2\text{ k} + 1.2\text{ k}) \approx 4\text{ V}$$
 [Ans.]

Using the same equivalents for transistor Q_2 we can find,

$$V_{G2} = V_{D1} = V_{DD} - I_{D1} R_{D1} = 12\text{ V} - 5.87\text{ mA} \times 1.2\text{ k} \approx 5.1\text{ V}$$

$$V_{GS2} = V_{G2} + |V_{SS}| - I_{D2} R_{S2} = 5.1\text{ V} + 12\text{ V} - 2.2\text{ k} \times I_{D2}$$

Solving this equation, we get $I_{D2} = 8.02\text{ mA}$

$$V_{DS2} = 12\text{ V} + 12\text{ V} - 8.02\text{ mA}(2.2\text{ k} + .3\text{ k}) \approx 3.95\text{ V}$$

Therefore, the operating points of the transistors are,

Q_1 :
$$I_{D1} = 5.87\text{ mA}, \quad V_{DS1} = 4\text{ V}$$
 [Ans.]

Q_2 :
$$I_{D2} = 8.02\text{ mA}, \quad V_{DS2} = 3.95\text{ V}$$
 [Ans.]

The transconductance, g_m can be calculated using Equ.(7-12).

$$g_{m1} = \frac{2\sqrt{I_{DSS1}I_{D1}}}{|V_{P1}|} = \frac{2\sqrt{12 \text{ mA} \times 5.87 \text{ mA}}}{|3 \text{ V}|} \approx 6.6 \text{ mS}$$

and,

$$g_{m2} = \frac{2\sqrt{I_{DSS2}I_{D2}}}{|V_{P2}|} = \frac{2\sqrt{12 \text{ mA} \times 8.02 \text{ mA}}}{|3 \text{ V}|} \approx 6.54 \text{ mS}$$

Voltage gain of the first stage can be calculated using Equ.(12-51),

$$A_{v1} = -g_{m1}(r_{d1} \parallel R_{D1} \parallel R_{G2} \parallel R_{gs2})$$

There is no R_{G2} for second stage and R_{gs2} is very large. Therefore,

$$A_{v1} = -g_{m1}(r_{d1} \parallel R_{D1}) = -6.6 \text{ mS} (200 \text{ k} \parallel 1.2 \text{ k}) \approx -7.87$$

Voltage gain of the second stage can be calculated using the same equation,

$$A_{v2} = -g_{m2}(r_{d2} \parallel R_{D2} \parallel R_L) = -6.54 \text{ mS} (200 \text{ k} \parallel 0.5 \text{ k} \parallel 4.7 \text{ k})$$

or,

$$A_{v2} \approx -3$$

The overall voltage gain of the multistage amplifier will be the product of the individual voltage gain. Therefore,

$$A_v \approx A_{v1} \times A_{v2} = (-7.87) \times (-3) \approx 23.6$$

Comments: Gain of FET amplifier is lower than BJT amplifiers. Here, $V_{G1} = 0 \text{ V}$, so the source can be connected directly without using C_{ic} .

12.14 BiFET Amplifier

So far we have found that, amplifiers can be designed using both BJTs and FETs. We know that, BJT amplifiers produce very high voltage gain but suffer from low or very low input impedance (e.g. CB amplifiers). On the other hand, FET amplifiers have very high input impedance (e.g. CS amplifiers) but their voltage gain is comparatively low. Even a two stage FET amplifier cannot produce voltage gain like a single stage BJT amplifier. To achieve the advantages of FET and BJT, sometimes they are combinedly used in a multistage amplifier. This amplifier is called **BiFET amplifier**. To utilize the high impedance property the FET is used at the first stage, and to get high amplification next one or two stages are made of BJTs. Although the gain of an n-stage BiFET amplifier is lower than that of an n-stage BJT amplifier, BiFET is a trade-off between gain and input impedance. The voltage gain of a multistage BiFET amplifier is much better than that of a FET amplifier consisting of same number of stages. Three types of coupling methods described so far can also used in BiFET amplifiers.

12.15 RC Coupled BiFET Amplifier

A two-stage, RC coupled BiFET amplifier is shown in Fig.12-20. The first stage is a CS FET amplifier with voltage divider bias and the second stage is a CE BJT amplifier with the same voltage divider bias. Depending on the design, the first stage will produce a voltage gain of approximately 2 to 20 but it will provide very high input impedance. We have already found that for CS amplifier, $Z_i = R_G$

and its value depends on bias resistors R_1 and R_2 . The minimum value of Z_i can be set approximately 100 k Ω . On the other hand, the input impedance of CE amplifier is $Z_i = h_{ie2}$ and its maximum value may be 5 k Ω . But with proper design, the voltage gain of the CE amplifier can be made as high as 200.

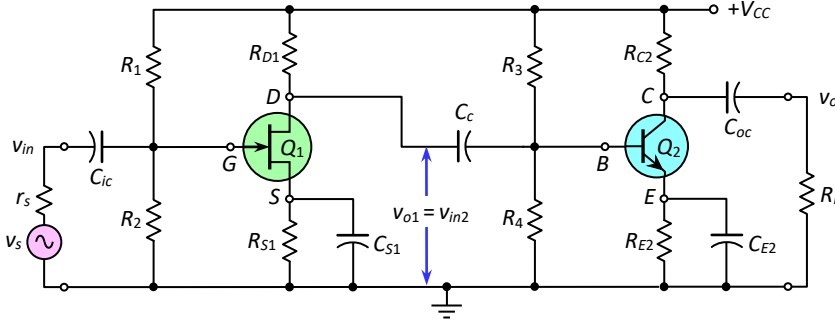


Fig.12-20: Two-stage RC coupled BiFET amplifier

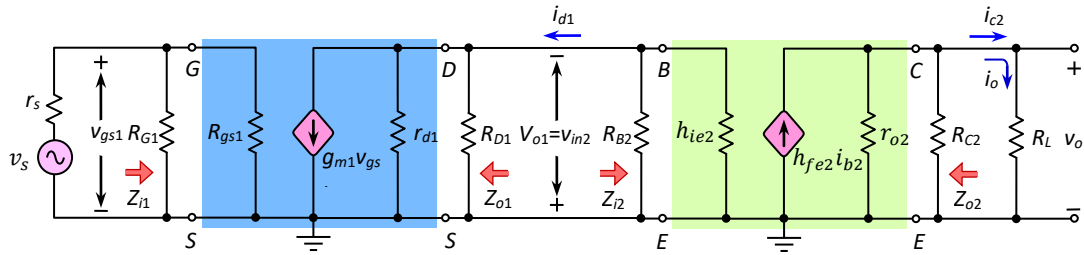


Fig.12-21: AC equivalent circuit of two-stage RC coupled BiFET amplifier

The AC equivalent circuit of this BiFET amplifier is shown in Fig.12-21. From this equivalent circuit and the knowledge of previous study, we can find that the voltage gain of the CS amplifier is,

$$A_{v1} = -g_{m1}(r_{d1} \parallel R_{D1} \parallel R_{C2} \parallel h_{ie2}) \quad (12-57)$$

$\because r_{d1}$ is very large,

$$A_{v1} = -g_{m1}(R_{D1} \parallel R_{C2} \parallel h_{ie2}) \quad (12-58)$$

Similarly, the voltage gain of the second stage is,

$$A_{v2} = -f_{fe2}(r_{o2} \parallel R_{C2} \parallel R_L) \quad (12-59)$$

$\because r_{o2}$ is very large,

$$A_{v2} = -f_{fe2}(R_{C2} \parallel R_L) \quad (12-60)$$

The overall voltage gain of the multistage amplifier will be the product of the individual voltage gain. Therefore,

$$A_v \approx A_{v1} \times A_{v2} = g_{m1}(R_{D1} \parallel R_{C2} \parallel h_{ie2}) \times f_{fe2}(R_{C2} \parallel R_L) \quad (12-61)$$

Example 12-11

Design a small signal RC coupled BiFET amplifier to achieve voltage gain as much as possible assuming $V_{CC} = 20$ V, to amplify a signal with maximum amplitude ± 0.1 mV. Consider a transistor

with $h_{fe(min)} = 75$, and a JFET with $V_{P(max)} = -3 \text{ V}$, and $I_{DSS(max)} = 12 \text{ mA}$, $r_s = 600 \Omega$, $R_L = 5.6 \text{ k}$, and $f_{L(m)} = 100 \text{ Hz}$.

Solution:

Design of the first stage:

Let, $I_{D(max)} = 5 \text{ mA}$.

From Shockley's equation,

$$V_{GS} = V_P(1 - \sqrt{I_D/I_{DSS}}) = -3(1 - \sqrt{5/12}) = -1.1 \text{ V}$$

Minimum drain-source voltage will be [Equ.()] **[H_C_atten]**,

$$V_{DS(min)} = |V_{P(max)}| + 1 \text{ V} - |V_{GS}| = 3 + 1 - 1.1 \approx 2.9 \text{ V}$$

To get higher gain we have to assign less voltage to R_S and more voltage to R_D . Let $V_{RS} = 7.5 \text{ V}$ and $V_{RD} = V_{DD} - V_S - V_{DS(min)} = 20 \text{ V} - 7.5 \text{ V} - 2.9 \text{ V} = 9.6 \text{ V}$.

$$\therefore R_{D1} = \frac{V_{RD}}{I_D} = \frac{9.6 \text{ V}}{5 \text{ mA}} = 1.92 \text{ k, use standard } 1.8 \text{ k}$$

$$R_{S1} = \frac{V_{RD}}{I_D} = \frac{7.5 \text{ V}}{5 \text{ mA}} = 1.5 \text{ k (standard)}$$

The required gate voltage will be,

$$V_G = V_S - |V_{GS}| = 7.5 \text{ V} - 1.1 \text{ V} = 6.4 \text{ V}$$

Let $R_1 = 1 \text{ M}$. Therefore,

$$R_2 = R_1 \times \frac{V_G}{V_{DD} - V_G} = 1 \text{ M} \times \frac{6.4 \text{ V}}{20 - 6.4 \text{ V}} = 471 \text{ k, use standard } 470 \text{ k}$$

Design of the second stage:

Let $I_C = 1 \text{ mA}$ and $V_{E2} = 5 \text{ V}$, $V_{CE2} = 2 \text{ V}$. So, $V_{RC2} = V_{CC} - V_{E2} - V_{CE} = 20 \text{ V} - 5 \text{ V} - 2 \text{ V} = 13 \text{ V}$.

$$\therefore R_{C2} = \frac{V_{RC2}}{I_{D2}} = \frac{13 \text{ V}}{1 \text{ mA}} = 13 \text{ k (standard)}$$

$$\text{and, } R_{E2} \approx \frac{V_{E2}}{I_{C2}} = \frac{5 \text{ V}}{1 \text{ mA}} = 5 \text{ k, use standard } 5.1 \text{ k}$$

Base voltage will $R_{E2} \times I_{E2} + 0.7 \text{ V} = 5.1 \text{ k} \times 1 \text{ mA} + 0.7 \text{ V} = 5.8 \text{ V}$.

Let the current flowing through R_4 be $I_{C2}/10 = 0.1 \text{ mA}$.

$$\therefore R_4 = \frac{V_{R4}}{I_{R4}} = \frac{5.8 \text{ V}}{0.1 \text{ mA}} = 58 \text{ k, use } 56 \text{ k and } 2 \text{ k}$$

Base current will be, $I_{B2} = I_{C2}/h_{fe(min)} = 1 \text{ mA}/75 = 0.013 \text{ mA}$.

Therefore, current flowing through R_3 will be $0.1 \text{ mA} + 0.013 \text{ mA} = 0.1013 \text{ mA}$.

$$\therefore R_3 = \frac{V_{R3}}{I_{R3}} = \frac{V_{CC} - V_{B2}}{I_{R3}} = \frac{20\text{ V} - 5.8\text{ V}}{0.1013\text{ mA}} \approx 140\text{ k}, \text{ use } 110\text{ k and } 30\text{ k}$$

Values of all coupling capacitors can be calculated using the similar process as in Example 12-12.

The value of inter-stage coupling capacitor can be calculated using Equ.(12-17) using R_{D1} in place of R_{C1} ,

$$C_c = \frac{10}{2\pi f_{L(m)}(R_{D1} + R_{B2} || h_{ie2})}$$

Here, $h_{ie2} = h_{fe2} \times 26\text{ mV}/I_{C1} = 75 \times 26\text{ mV}/1\text{ mA} = 1.95\text{ k}$

and, $R_{B2} = R_3 || R_4 = 140\text{ k} || 58\text{ k} \approx 41\text{ k}$

$$\therefore C_c = \frac{10}{2\pi 100(1.8\text{ k} + 41\text{ k} || 1.95\text{ k})} \approx 5\text{ }\mu\text{F}, \text{ use standard } 5.6\text{ }\mu\text{F}$$

If we calculate the overall gain of this amplifier, we find that $A_{vm} = (-9.464) \times (-500) = 4732$. So the swing at the collector of Q_2 will be $A_{vm} \times v_{in(max)} = 4732 \times \pm 0.1\text{ mV} = 0.4732\text{ V}$. As we have set $V_{CE2} = 2\text{ V}$, the amplifier will work properly.

Comments: To maximize the gain, we have assigned a very large voltage across the collector resistor that results in a large value of R_C .

12.16 Direct Coupled BiFET Amplifier

A direct coupled BiFET amplifier is shown in Fig.12-22. In this circuit, the first stage is a CS amplifier that employs an N-channel JFET, and the second stage is a CE amplifier that employs a PNP transistor. We know that in direct coupled multistage amplifier, there is no DC isolation. That is, the drain voltage of first stage will be the base voltage of the second stage. In FET amplifiers, we have to assign larger voltage to source compared to the voltage assigned to the emitter of a BJT amplifier. Therefore, the drain voltage (V_{D1}) of Q_1 will be higher than voltage across the drain resistor R_D . That is V_{D1} will be more than 50% of supply voltage (V_{CC}). Now consider a direct coupled BiFET amplifier with NPN transistor (imagine Fig.12-20 without R_1 , R_4 and C_{oc} short). For this case V_{B2} will be equal to V_{D1} and V_{E2} will be ($V_{B2} - 0.7\text{ V}$). Thus more than 50% supply voltage will go to emitter. That results in a very small voltage across R_{C2} . Small voltage across R_{C2} results in small collector resistance (R_C) and hence small voltage gain. On the contrary, if we use PNP transistor, more than 50% voltage will be allocated for V_{CE} and V_{RC2} and less voltage will be allocated to the emitter of Q_2 . This is the advantage of using PNP transistor in direct coupled BiFET amplifiers.

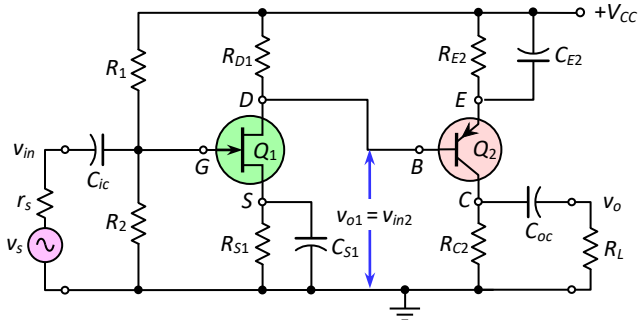


Fig.12-22: Two-stage direct coupled BiFET amplifier

Using the concept of RC coupled BJT amplifier and the equivalent circuit of CS and CE amplifier we can find the voltage gain the individual stages. The voltage gain of the CS amplifier will be,

$$A_{v1} = -g_{m1}(r_{d1} \parallel R_{D1} \parallel h_{ie2})$$

$\because r_{d1}$ is very large,

$$(A_{v1} = -g_{m1}(R_{D1} \parallel R_{C2} \parallel h_{ie2})) \quad (12-62)$$

Voltage gain of the second stage will be,

$$A_{v2} = -f_{fe2}(r_{o2} \parallel R_{C2} \parallel R_L)$$

If r_{o2} is very large,

$$(A_{v2} = -f_{fe2}(R_{C2} \parallel R_L)) \quad (12-63)$$

The overall voltage gain of the multistage amplifier will be the product of the individual voltage gain. Therefore,

$$A_v \approx A_{v1} \times A_{v2} = g_{m1}(R_{D1} \parallel R_{C2} \parallel h_{ie2}) \times f_{fe2}(R_{C2} \parallel R_L) \quad (12-64)$$

Example 12-12

Design a two-stage BiFET amplifier like the circuit of Fig.12-22. Assume $h_{fe2} = 100$, $I_{DSS1} = 12 \text{ mA}$, $V_P = -3 \text{ V}$, $f_{L(m)} = 100 \text{ Hz}$, $R_L = 4.7 \text{ k}$, $r_s = 0.6 \text{ k}$, and $V_{CC} = 16 \text{ V}$.

Solution:

Let, $I_{D1} = 4 \text{ mA}$, and $I_{C2} = 2 \text{ mA}$.

Using Shockley's equation,

$$V_{GS} = V_P(1 - \sqrt{I_D/I_{DSS}}) = -3(1 - \sqrt{4/12}) = -1.268 \text{ V}$$

Minimum drain-source voltage will be,

$$V_{DS(min)} = |V_P| + 1 \text{ V} - |V_{GS}| = (3 + 1 - 1.268) \text{ V} = 2.732 \text{ V}$$

Let, the rest of the voltage equally divide between R_{D1} , and R_{S1} .

$$V_{RD1} = V_{RS1} = \frac{1}{2}(V_{DD} - V_{DS(min)}) = \frac{1}{2}(16 \text{ V} - 2.732 \text{ V}) \approx 6.634 \text{ V}$$

$$\therefore V_{RD1} = V_{RS1} = \frac{6.634 \text{ V}}{4 \text{ mA}} = 1.66 \text{ k}, \text{ use standard } 1.5 \text{ k}$$

Actual V_{RD1} , and V_{RS1} will be $1.5 \text{ k} \times 4 \text{ mA} = 6 \text{ V}$.

Required gate voltage will be,

$$V_{G1} = I_D R_{S1} - |V_{GS1}| = 4 \text{ mA} \times 1.5 \text{ k} - 1.268 \text{ V} = 4.732 \text{ V}$$

Let $R_1 = 1 \text{ M}$. Therefore,

$$\frac{R_2}{R_1} = \frac{V_{G1}}{V_{DD} - V_{G1}} = \frac{4.732 \text{ V}}{16 \text{ V} - 4.732 \text{ V}} \approx 0.42$$

$$\therefore R_2 = 0.42 \times R_1 = 0.42 \times 1 \text{ M} = 420 \text{ k}, \text{ use } 300 \text{ k} + 120 \text{ k}$$

Drain voltage of Q_1 which is also the base voltage of Q_2 will be,

$$V_{D1} = V_{B2} = V_{DD} - I_{D1}R_{D1} = 16 \text{ V} - 6 \text{ V} = 10 \text{ V}$$

$$\therefore V_{E2} = V_{B2} + V_{BE2} = 10 \text{ V} + 0.7 \text{ V} = 10.7 \text{ V}$$

$$\therefore V_{RE2} = V_{CC} - V_{E2} = 16 \text{ V} - 10.7 \text{ V} = 5.3 \text{ V}$$

$$\therefore R_{E2} = \frac{V_{E2}}{I_{C2}} = \frac{5.3 \text{ V}}{2 \text{ mA}} = 2.65 \text{ k}, \text{ use standard } 2.7 \text{ k}$$

Let, $V_{CE(min)} = 3 \text{ V}$. So $V_{RC2} = V_{CC} - V_{RE2} - V_{CE2(min)}$

$$\therefore V_{RC2} = V_{CC} - V_{RE2} - V_{CE2(min)} = 16 \text{ V} - 5.3 \text{ V} - 3 \text{ V} = 7.7 \text{ V}$$

$$\therefore R_{C2} = \frac{V_{RC2}}{I_{C2}} = \frac{7.7 \text{ V}}{2 \text{ mA}} = 3.85 \text{ k}, \text{ use standard } 3.9 \text{ k}$$

Calculation of the Capacitances:

Using Equ.(12-9-31)

$$C_{ic} = \frac{10}{2\pi f_L (r_s + Z_i)}$$

Here,

$$Z_i \approx R_{G1} = R_1 \parallel R_2 = 1 \text{ M} \parallel 420 \text{ k} \approx 295.78 \text{ k}$$

$$\therefore C_{ic} = \frac{10}{2\pi \times 100(0.6 \text{ k} + 295.78 \text{ k})} \approx 5.4 \text{ } \mu\text{F}, \text{ use standard } 5.6 \text{ } \mu\text{F}$$

Using Equ.(12-9-20)

$$C_{oc} = \frac{10}{2\pi f_L (R_{C2} + R_L)} = \frac{10}{2\pi \times 100(3.9 \text{ k} + 4.7 \text{ k})} \approx 1.9 \text{ } \mu\text{F} \text{ (} 2.2 \text{ } \mu\text{F)}$$

or,

$$C_{oc} = \frac{10}{2\pi \times 100(3.9 \text{ k} + 4.7 \text{ k})} \approx 1.9 \text{ } \mu\text{F}, \text{ use standard } 2.2 \text{ } \mu\text{F}$$

Using Equ.(12-12),

$$C_{E2} = \frac{1 + h_{fe2}}{2\pi [0.65 f_{L(m)}] h_{ie2}}$$

Here,

$$h_{ie2} = h_{fe2} \frac{26 \text{ mV}}{I_{C2}} = 100 \times \frac{26 \text{ mV}}{2 \text{ mA}} = 2.6 \text{ k}$$

$$\therefore C_{E2} = \frac{1 + 100}{2\pi [0.65 \times 100] 2.6 \text{ k}} \approx 95 \text{ } \mu\text{F}, \text{ use standard } 100 \text{ } \mu\text{F}$$

Using Equ.(12-12),

$$C_{S1} = \frac{g_{m1}}{2\pi \times 0.65 \times f_{L(m)}}$$

Using Equ.(12-6-4)

$$g_{m1} = \frac{2\sqrt{I_{DSS1} I_{D1}}}{|V_{P1}|} = \frac{2\sqrt{12 \text{ mA} \times 4 \text{ mA}}}{|3 \text{ V}|} \approx 4.6 \text{ mS}$$

$$\therefore C_{S1} = \frac{4.6 \text{ mS}}{2\pi \times 0.65 \times 100} \approx 11 \text{ } \mu\text{F}, \text{ use standard } 10 \text{ } \mu\text{F}$$

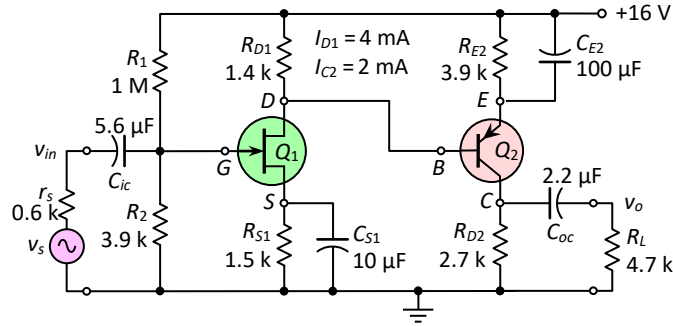


Fig.12-23: BiFET direct coupled amplifier designed in Example 12-12

Answer: The designed amplifier is shown in Fig.12-23.

Comments: To get more gain, we can decrease the value of I_{C2} and can assign more voltage to R_C resistor.

12.17 BiFET Cascode Amplifier

As found in Example 12-, due to the small input impedance of CB amplifier (in second stage) the gain of the first stage CE is very low. This is a common case of cascade amplifier. As the high voltage gain facility of CE amplifier cannot be used here, we can replace this CE amplifier by a CS amplifier. This FET amplifier will give low voltage gain but very large input impedance. The resultant two stage amplifier will be a *BiFET cascade* amplifier as shown in Fig.12-.

The AC equivalent circuit of this amplifier is shown in Fig.12-23.

Voltage gain of the first stage will be,

$$A_{v1} = -g_{m1}(r_{d1} \parallel R_{D1} \parallel h_{ib2})$$

$\because r_{d1}$ is very large,

$$\boxed{A_{v1} = -g_{m1}(R_{D1} \parallel h_{ib2})} \quad (12-65)$$

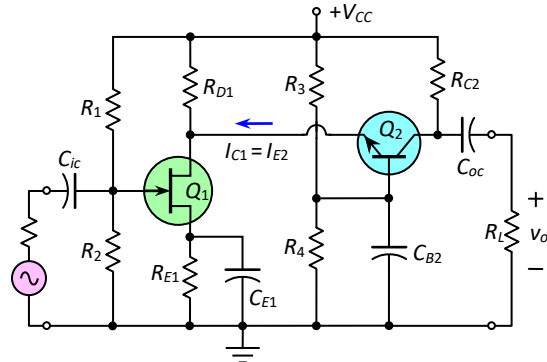


Fig.12-24: BiFET Cascode amplifier

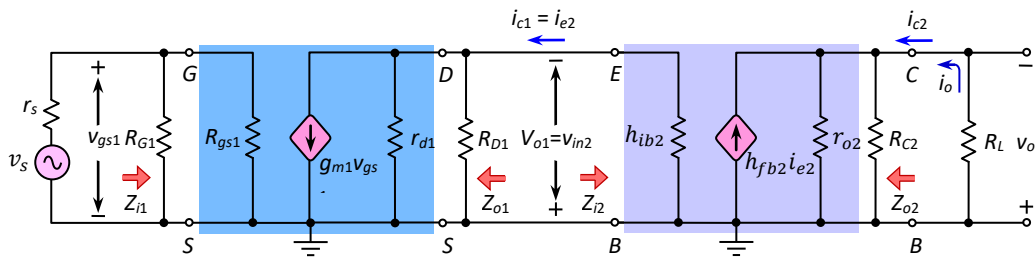


Fig.12-25: AC equivalent circuit of BiFET cascode amplifier

Voltage gain of the second stage will be,

$$A_{v2} = \frac{h_{fb2}(r_{o2} \parallel R_{C2} \parallel R_L)}{h_{ib2}}$$

$\because r_o$ is very large,

$$A_{v2} \approx \frac{h_{fb2} R_{C2} \parallel R_L}{h_{ib2}} \quad (12-66)$$

The overall voltage gain of this multistage amplifier will be the product of the individual voltage gain. Therefore,

$$A_{v(m)} = A_{v1} \times A_{v2} \approx -g_{m1}(r_{d1} \parallel R_{D1} \parallel h_{ib2}) \times \frac{h_{fb2} R_{C2} \parallel R_L}{h_{ib1}} \quad (12-67)$$

12.18 Sample Questions

1. What is a multistage amplifier and why we use the multistage amplifier?
2. Prove that the overall voltage gain of a multistage amplifier is the product of the voltage gain of the individual amplifiers.
3. What do you mean by loading effect? Explain how the gain of an amplifier is affected when used in cascaded form.
4. Explain how the bandwidth of a multistage amplifier is determined. What will be the effective lower and higher cut-off frequencies of a multistage amplifier if all the stages have the same lower and higher cut-off frequencies?
5. What are the main coupling methods used in multistage amplifiers?
6. Draw the circuit diagram of a two stage CE amplifier using RC coupling method and describe.
7. Draw the equivalent circuit diagram of a two stage CE amplifier with RC coupling and describe estimate its overall voltage gain.
8. What are the advantages and disadvantages of RC coupling method used in multistage amplifiers?
9. Explain how to determine the coupling and bypass capacitors used in the RC coupled multistage amplifiers?
10. What are the main advantages and disadvantages of direct coupling method used in multistage amplifiers?

11. Draw the circuit diagram of a two stage CE amplifier using direct coupling method and describe.
12. Draw the equivalent circuit diagram of a two stage CE amplifier with direct coupling and describe estimate its overall voltage gain.
13. Why we use Darlington transistor in multistage amplifier? Draw the circuit diagram of a two stage amplifier with a Darlington transistor and describe. Calculate the overall voltage gain of this amplifier.
14. What are the applications of transformer coupled amplifiers?
15. What are the main advantages and disadvantages of transformer coupling method used in multistage amplifiers?
16. Draw the circuit diagram of a two stage transformer coupled CE amplifier and describe.
17. Draw the equivalent circuit diagram of a two stage CE amplifier with transformer coupling and estimate its overall voltage gain.
18. Draw the typical frequency response of a transformer coupled multistage amplifier and describe.
19. Explain the function of the bypass capacitor C_{B2} used in the circuit of Fig.12-11. How the capacitance of this amplifier can be determined?
20. Give a comparison among the three types of coupling methods used in multistage amplifier.
21. Why we use cascade amplifier? Draw the circuit diagram of a cascade amplifier and describe.
22. Draw the equivalent circuit of a cascade amplifier and derive the expression for the voltage gain.
23. Draw the circuit diagram of a two stage RC coupled CS amplifier and describe.
24. Draw the equivalent circuit diagram of an RC coupled two stage CS and calculate its overall voltage gain.
25. Draw the circuit diagram of a two stage direct coupled CS amplifier and describe.
26. Draw the equivalent circuit diagram of a direct coupled two stage CS and calculate its overall voltage gain.
27. What do you mean by BiFET amplifier? What are the advantages of this amplifier?
28. Draw the circuit diagram of a two stage BiFET amplifier using RC coupling method and describe.
29. Draw the equivalent circuit diagram of a two stage CE amplifier with RC coupling and describe estimate its overall voltage gain.
30. Why is a PNP transistor used in a direct coupled BiFET amplifier- Explain with circuit diagram.
31. Calculate the voltage gain of a two stage direct coupled BiFET amplifier.
32. What is BiFET cascade amplifier? Draw the circuit diagram of a BiFET cascade amplifier and describe its operation.
33. What will be the overall voltage gain of a BiFET cascade amplifier?
34. What are the main advantages of a BiFET cascade amplifier?
35. Mention some applications of multistage amplifiers.

12.19 Exercise

1. A four-stage amplifier is constructed using four CE amplifier as shown in Fig.12-2. The voltage gain of the individual amplifiers are given as: $A_{v1} = -10$, $A_{v2} = -12$, $A_{v3} = 18$, and $A_{v4} = -13$. Calculate the overall voltage gain and phase of the cascaded amplifier phase. What will be the overall gain in dB scale?
2. Calculate the voltage gain of the multistage amplifier of Ex.1 considering the loading effect. Calculate its value in dB scale, as well. Assume, $Z_{i1} = 1.1 \text{ k}$, $Z_{o1} = 2.3 \text{ k}$, $Z_{i2} = 3.2 \text{ k}$, $Z_{o2} = 2.5 \text{ k}$, $Z_{i3} = 3.3 \text{ k}$, $Z_{o3} = 1.5 \text{ k}$, and $Z_{i4} = 2.5 \text{ k}$, $Z_{o4} = 1.2 \text{ k}$. The source internal resistor is 600Ω and the load resistance is 3.3 k . What will be the value of the load voltage if the source voltage of $v_s = 10 \text{ mV} \sin \omega t$.
3. A multistage amplifier is implemented by cascading three CE amplifiers. The lower and higher cut-off frequencies of the amplifiers are: $f_{L1} = 300 \text{ Hz}$, $f_{H1} = 5.0 \text{ MHz}$, $f_{L2} = 310 \text{ Hz}$, $f_{H2} = 5.1 \text{ MHz}$ and $f_{L3} = 315 \text{ Hz}$, $f_{H3} = 5.12 \text{ MHz}$, respectively. Calculate the overall lower and higher cut-off frequencies of the multistage amplifier. Repeat the problem of Ex.3 for $f_{L1} = 100 \text{ Hz}$, $f_{H1} = 5.1 \text{ MHz}$, $f_{L2} = 105 \text{ Hz}$, $f_{H2} = 5.05 \text{ MHz}$ and $f_{L3} = 300 \text{ Hz}$, $f_{H3} = 2.2 \text{ MHz}$, respectively.
4. Fig.12-26 shows a multistage CE amplifier with voltage divider bias. The transistors, used in this circuit, have $h_{ie} = 1.2 \text{ k}$, $h_{fe} = 120$, and $r_o = 100 \text{ k}$. Calculate A_{v1} , A_{v2} , Z_{im} , A_{vm} , and Z_{om} . Calculate the voltage gain and lower cut-off frequency of individual amplifiers and those of the multistage amplifier.

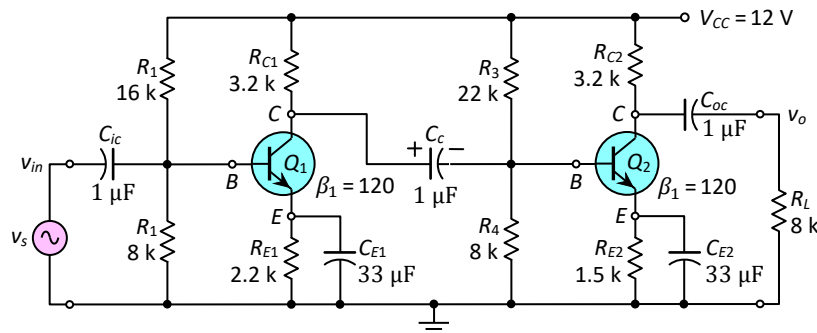


Fig.12-26: Two-stage RC coupled CE amplifier

5. Design an RC coupled two-stage amplifier with supply voltage of 16 V , and transistor having $h_{fe} = 120$, and $h_{ie} = 1.2 \text{ k}\Omega$. Assume $R_L = 5.2 \text{ k}$, $r_s = 600 \Omega$, and $f_L = 200 \text{ Hz}$. The voltage gain of the individual unit should be 50 or higher without loading effect.
6. Design a direct coupled multistage amplifier with supply voltage of 10 V , and transistor having $h_{fe} = 110$, and $h_{ie} = 1.1 \text{ k}\Omega$. Assume $R_L = 7.4 \text{ k}$, $r_s = 600 \Omega$, and $f_L = 200 \text{ Hz}$.
7. The transistors used in the multistage amplifier of Fig.12-27, have $\beta_1 = 150$, $\beta_2 = 120$, and $\beta_3 = 100$. Calculate A_v , A_i , Z_i , and Z_o of individual stages and the multistage amplifier.

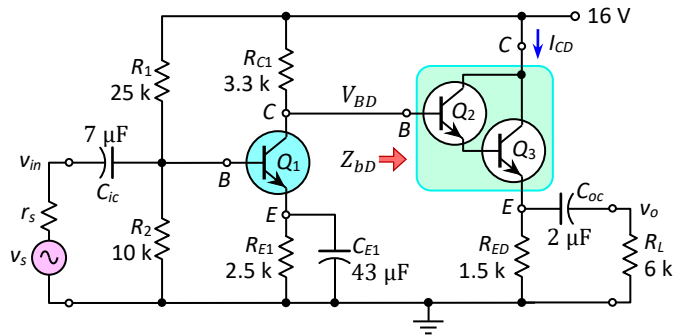


Fig.12-27: Two-stage direct coupled amplifier consisting of a CE amplifier and a CC amplifier with Darlington pair

8. Assume that the transistors used in the transformer coupled amplifier of Fig.12-28, have $h_{fe} = 80$, and $r_o = 100 \text{ k}$, and the resistance of the DC resistance of the primary coil is $R_{p1} = R_{p2} = 20 \Omega$. Calculate the DC operating points of the transistors, voltage gain of individual stages, and overall voltage gain v_o/v_s of the two-stage amplifier.

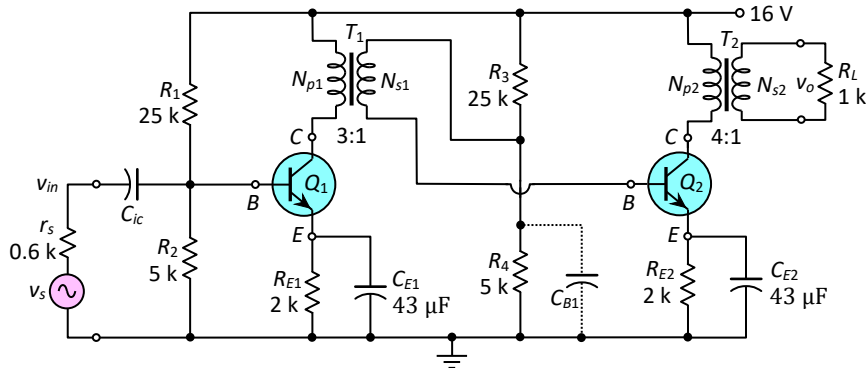


Fig.12-28: Two-stage transformer coupled CE amplifier

9. The transistors used in the cascode amplifier of Fig.12-29 have $h_{fe} = 120$, $r_o = 200 \text{ k}$, $C_{be} = 12 \text{ pF}$, $C_{bc} = 6 \text{ pF}$, and $C_{ce} = 2.5 \text{ pF}$. Calculate the following:
- DC operating points of individual transistors
 - Voltage gain of individual amplifier stages
 - Overall voltage gain of the cascode amplifier
 - Lower cut-off frequency and
 - The higher cut-off frequency of the cascode amplifier.

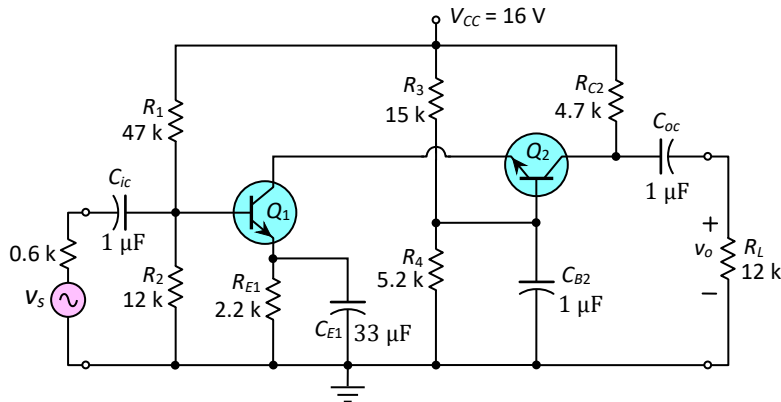


Fig.12-29: Cascode amplifier designed in Example 12-8

10. The JFETs used in the multistage amplifier of Fig.12-30 has $V_P = -4$ V, and $I_{DSS} = 12$ mA. Determine the DC operating point and voltage gain of the individual stages of this amplifier. Also determine the overall voltage gain of the multistage amplifier.

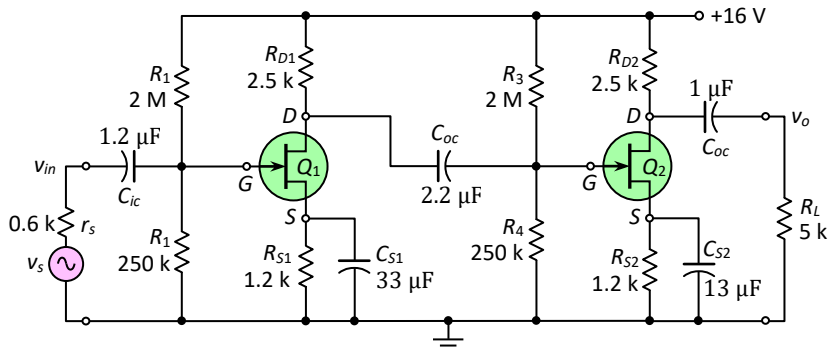


Fig.12-30: Two-stage RC coupled CS amplifier

11. For the CS multistage amplifier of Fig.12-30, calculate the lower cut-off and higher cut-off frequency. Also calculate its bandwidth. Assume, $C_{gd} = 3$ pF, $C_{gs} = 8$ pF, $C_{ds} = 1.5$ pF, the wiring capacitance is 1 pF on each side of the amplifiers, $g_m = 4.5$ mS, and $r_d = \infty$.
12. Suppose the two-stage direct coupled amplifier of Fig.12-31 uses supply voltage of ± 12 V, $R_1 = R_2 = R_3 = R_4 = 1$ M, $R_{S1} = R_{S2} = 2.2$ k, $R_{D1} = 1.2$ k, $R_{D2} = 300$ Ω, and $R_L = 4.7$ k. Assume that the JFETs have $I_{DSS} = -3$ V, $I_{DSS} = 12$ mA. Determine the operating points and voltage gain of the individual stages and the overall gain of the two-stage amplifier.

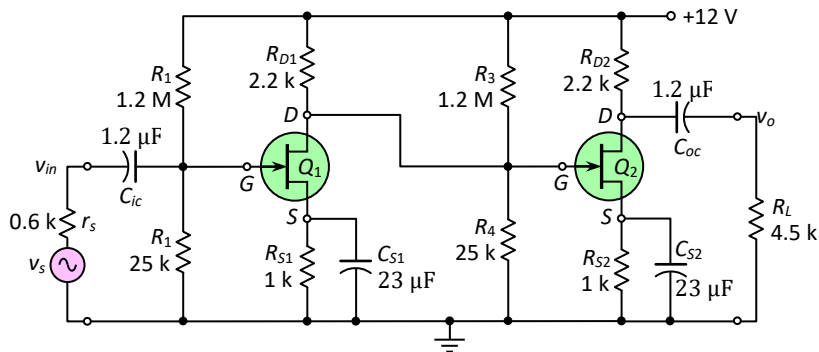


Fig.12-31: Two-stage RC coupled CS amplifier

13. Design a small signal RC coupled BiFET amplifier with $V_{CC} = 20$ V. Consider a transistor with $h_{fe(min)} = 70$, and a JFET with $V_{P(max)} = -4$ V, and $I_{DSS(max)} = 16$ mA, $r_s = 600 \Omega$, $R_L = 4.7$ k, and $f_{L(m)} = 200$ Hz. Can this amplifier amplify a signal with maximum amplitude ± 1 mV- if not why?
14. Design a two-stage BiFET amplifier like the circuit of Fig.12-22. Assume $h_{fe2} = 120$, $I_{DSS1} = 14$ mA, $V_P = -3.5$ V, $f_{L(m)} = 200$ Hz, $R_L = 6.5$ k, $r_s = 0.6$ k, and $V_{CC} = 20$ V.

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